

SECTION 2 – ROOM LAYOUTS

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2-1 INTRODUCTION

When laying out a floor plan there are special considerations that must be taken into account due to the magnetic field effect on equipment. The maximum magnetic field in which the upgrade supplied equipment can be located is listed in Table 2-1.

TABLE 2-1
PROXIMITY LIMITS

GAUSS (mT) LIMIT See Notes 1 & 3	EQUIPMENT	
0.5 GAUSS (0.05mT) OR LESS See Note 5	• Nuclear cameras	
1 GAUSS (0.1mT) OR LESS See Note 5	• Positron Emission Tomography scanner • Linear Accelerator • Cyclotrons • Accurate Measuring scale • Image intensifiers • Color TV	• Video display (color, B/W, monochrome) • CT scanner • Ultrasound • Lithotripter • Electron microscope • Advantage Workstation with CRT Monitor
3 GAUSS (0.3mT) OR LESS See Note 5	• Power transformers • Main electrical distribution transformers • Moving steel equipment such as: – Vehicular traffic – Fork lift trucks – Dumb waiters – Electric transport carts	– Loading dock (truck traffic) – Elevators – Escalators – Helicopters (See Note 4)
1 GAUSS (0.1mT) OR LESS	• Advantage Workstation with CRT Monitor	
10 GAUSS (1mT) OR LESS	• Operator Workspace Host Monitor (Color)	
30 GAUSS (3mT) OR LESS	• System Control Cabinet	• Heat Exchanger for Mobile System Gradient Coil Cooling
50 GAUSS (5mT) OR LESS	• Octane Computer • Operator Workspace Cabinet • LCD Color Monitor for OW (See note 6) • SRF Cabinet • ACGD/PDU Cabinet • Water Chiller for Gradient Coil Cooling • Main Disconnect Device	• Magnet Monitor • Telephones • Metal detector for screening • LCD Color Monitor for Advantage Workstation • MNS Cabinet
100 GAUSS (10mT) OR LESS	• Service Tool Magnet Power Supply Cabinet • Service Tool Shim Power Supply Cabinet	• Shield/Cryo Cooler Compressor Cabinet • Pneumatic Patient Alert Control Box
200 GAUSS (20mT) OR LESS	• Penetration Panel • Blower Box	• Magnet Rundown Unit
Note 1 Refer to SECTION 3, MAGNETIC FIELD CONSIDERATIONS, for magnet field plots. 2 Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable. 3 Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable. 4 Verify operating limits with provider of helicopter service 5 The 0.5 gauss (0.05 mT), 1 gauss (0.1 mT), and 3 gauss (0.3 mT) or less proximity limits apply to ALL magnets. For additional Active Shield magnet proximity limits refer to Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-12 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC 6 If gauss limit is more than indicated for OW with LCD Color Monitor then contact GE to determine impacts on OW equipment.		

2-2 TWO MAGNET SITE LAYOUT

Refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation*** Section 2-3 TWO MAGNET SITE LAYOUT for two magnet site layout information.

2-3 MINIMUM DOOR/HALLWAY SIZES

Table 2-2 lists minimum actual clearance opening dimensions for doors and hallways required by new equipment installed as part of the various Signa LX Upgrades. Installation or replacement of components listed in Table 2-3 must be taken into consideration when determining delivery route hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for component shipping dimensions.

TABLE 2-2
MINIMUM HALLWAY/DOOR DIMENSIONS

COMPONENT	MINIMUM HALLWAY/ DOOR WIDTH* in. (mm)	MINIMUM HALLWAY/ DOOR HEIGHT* in. (mm)	COMMENTS
Signa LX Operator Workspace Table	32 (813)	80 (2032)	
Signa LX Cabinets	36 (914)	80 (2032)	
Cryogen delivery route and Storage Room	43 (1092)	80 (2032)	Width requirements due to size of 500 liter dewars. Width and height requirements vary dependent on the dewars used. Check with cryogen supplier.
LCC Magnet	Refer to Note 1	Refer to Note 2	Refer to Table 2-3 for uncrated magnet dimensions.
RF Room Door	Refer to Section 7, RF SHIELDED ROOM	Refer to Section 7, RF SHIELDED ROOM	
Note * Minimum hallway and door dimensions are actual clearance openings. Width and height of rigging equipment is not included in above dimension. 1 Minimum width depends on access route to removable panels of RF shielded room wall. For straight path (i.e. no bends or turns) recommended to allow 6 in. (153 mm) on both sides of magnet. Appropriate calculations must be performed if turns exist along proposed magnet delivery route. Illustration NO TAG shows dimensions for 90° turn. 2 Final dimension is dependent on rigger equipment used, refer to <i>Direction 2223170 Signa MR/i & CV/i Pre-Installation</i> SECTION 8, SHIPPING AND DELIVERY DATA.			

2-3 MINIMUM DOOR/HALLWAY SIZES (Continued)

TABLE 2-3
UPGRADE COMPONENT DIMENSIONS FOR INSTALLATION/REPLACEMENT

COMPONENT	APPROXIMATE WEIGHT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	COMMENTS
LCC Magnet (uncrated)	Refer to comments.	Refer to comments.	Refer to Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 8, SHIPPING AND DELIVERY DATA, for dimensions, illustrations and weights.
Split Bridge (LCC Magnet Enclosure)	40 (18)	21.5 x 77.3, x 7 (546 x 1969 x 18)	Room dimensions in front of LCC Magnet MUST allow for bridge installation/servicing and Gradient Coil Replacement, See Note 3.
Replacement Gradient (BRM) Coil Assembly on a Shipping Cradle/Cart	See Note 1	35 x 96.09 x 55.88 (889 x 2441 x 1420) See Note 2 & 3	Initial BRM Gradient Coil Assembly is shipped installed in the Magnet. Shipping/installation cart is used to install replacement coil assembly only. Refer to Section 2-3-1, Gradient Coil Assembly, Cradle, & Cart Requirements. The Short Room Gradient Coil Cart must be used to replace the BRM when the distance in front of the magnet is less than 117.6 inch (2987).
Enclosure Covers & End Bells	See Comments	See Comments	Refer to Section 2-8 COMPONENT DIMENSIONS, for details of individual assemblies and applicable catalogs.
Rear Pedestal	30 (13.6)	17 x 43 x 33 (432 x 1092 x 838)	
Note 1 The replacement BRM Gradient Coil Assembly weight is approximately 2300 lbs (1045 kg), the shipping cradle is 125 lbs (57 kg), and the Gradient Coil Assembly shipping/installation cart weighs 855 lbs (389 kg). Therefore total shipping weight is 3280 lbs (1490 kg). The coil assembly outside diameter x length dimensions are 35.1 x 73.5 in. (892 x 1867 mm). 2 RF/Gradient (BRM) Coil Assembly and shipping cart dimensions are with cart in lowest position. Cart can be adjusted to maximum height of 61.88 in. (1572 mm). 3 LCC and Cx Magnets MUST USE special Gradient Coil Replacement Tool Kit for replacing the Gradient Coil Assembly. The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters. Utilization of Gradient Coil Replacement Tool requires 118 in. (2997 mm) clear space in front of magnet (same clearance as Bridge installation).			

2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements

Horizon upgrades will require the 55 cm Body Coil to be removed using the old 'blue' wooden coil cart.

■ Signa LX System without Magnet Upgrade

International sites upgrade shipments will have the RF/Gradient (BRM) Coil shipped on a wooden (disposable) cradle, see Illustration 2-1. A forklift or crane/hoist rated for 4000 lbs (1818 kg) is required to position the coil/cradle assembly onto the cart for installing the coil into the magnet, see Illustration 2-2. GE Service to provide the coil cart and accessories for International Sites, refer to Section 10-5 INSTALLATION EQUIPMENT.

USA sites ONLY upgrade shipments will receive the new RF/Gradient (BRM) coil on an aluminum (re-useable) cradle mated to coil cart with accessories. GE MR Manufacturing will provide cradle/cart which will be returned to MR Manufacturing for reuse.

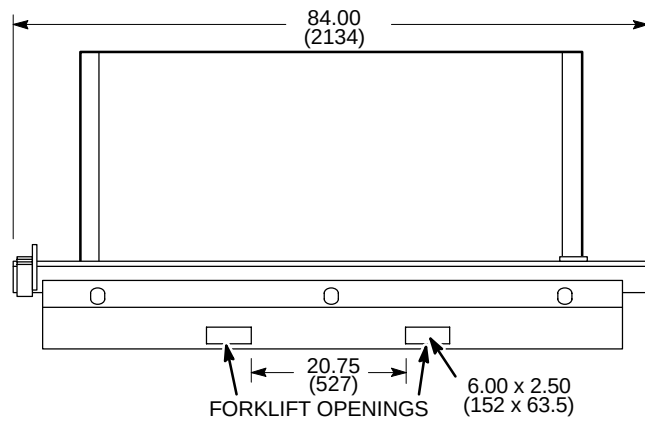
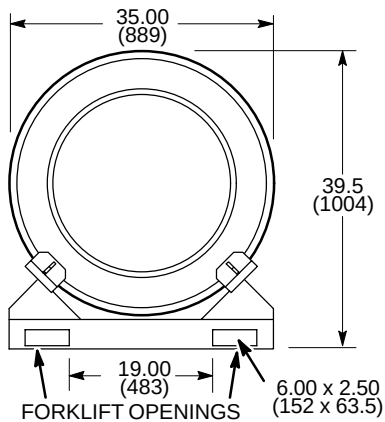
■ Systems Installing LCC Magnet Upgrade Concurrent With System Upgrade

Systems installing an LCC Magnet Upgrade at the same time as the System Upgrade will receive the LCC Magnet with the BRM Assembly installed inside the magnet bore and enclosure.

2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: WITH BRM 2425 lbs (1102 kg)

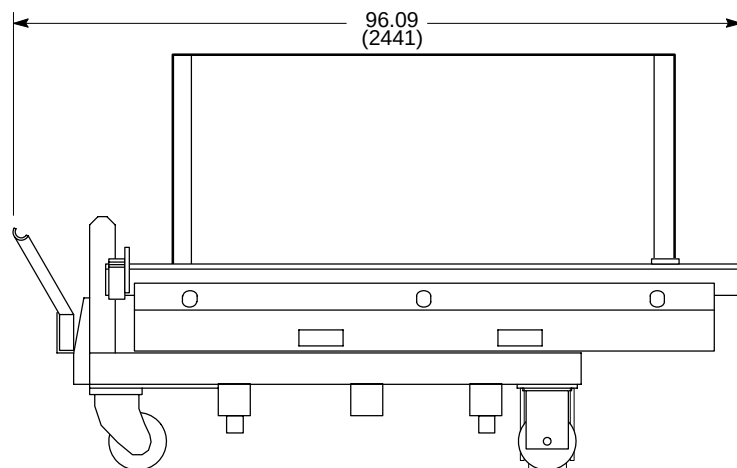
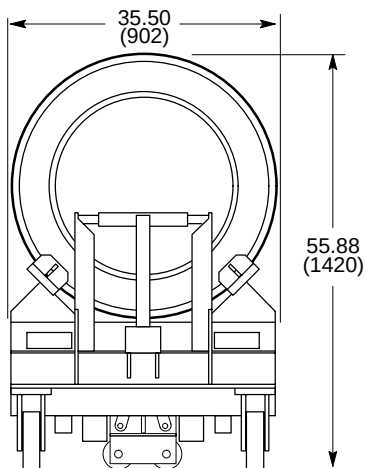


UPGRADE GRADIENT BRM COIL ASSEMBLY & CRADLE

ILLUSTRATION 2-1

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT WITH BRM 3280 lbs (1491 kg)



SHORT ROOM REPLACEMENT GRADIENT BRM COIL ASSEMBLY & CRADLE ON CART

ILLUSTRATION 2-2

2-4 CABLING CONSIDERATIONS

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes. Cabinets cable access areas are shown on illustrations in Section 2-8, COMPONENT DIMENSIONS. Some cable routing will be changed due to new equipment depending on system configuration being upgraded. Refer Section 6, INTERCONNECT DATA, for cables and their dimensions information.

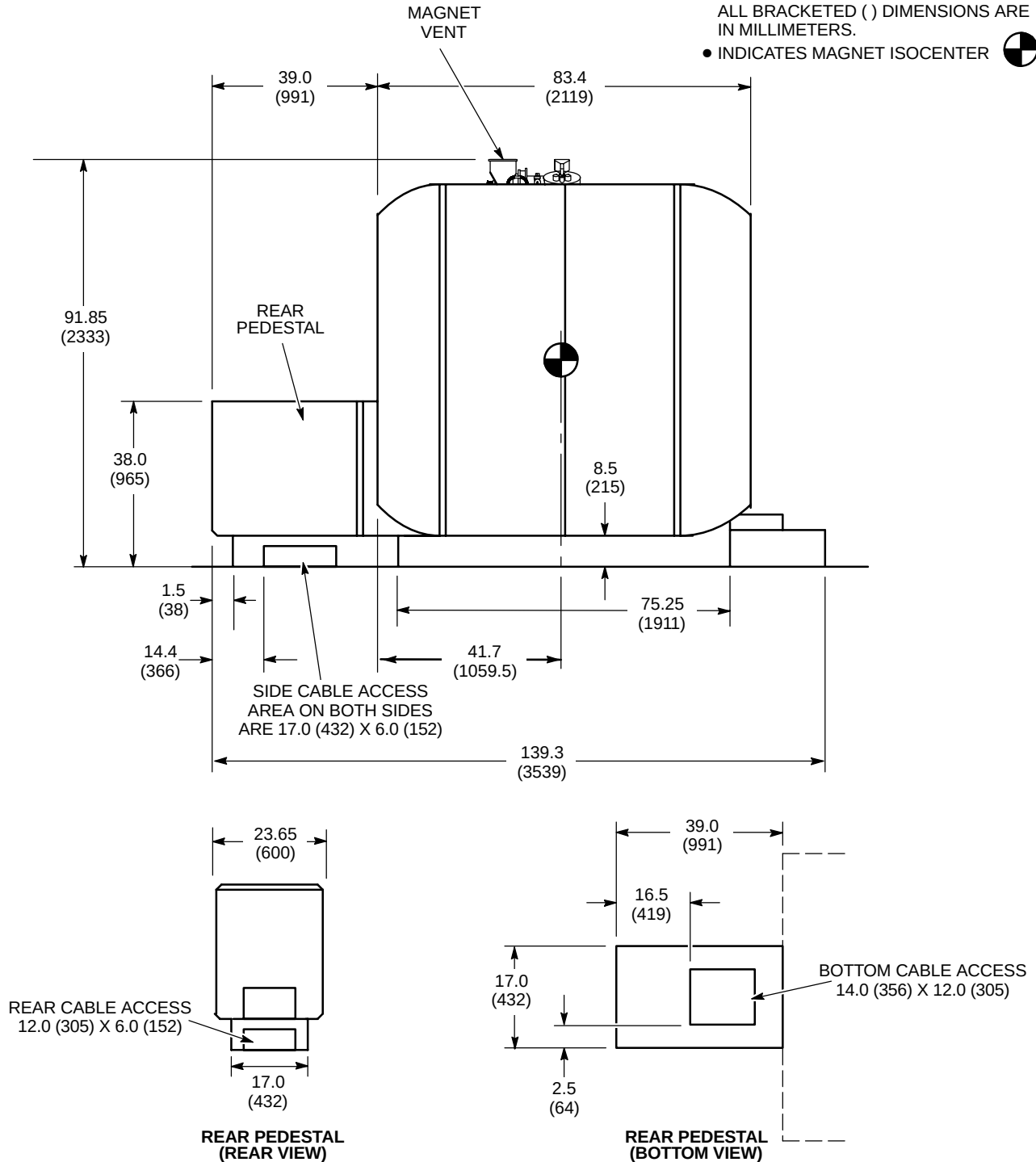
Note

Signa Horizon has an increase of cabling and hoses through the Rear Pedestal cable access areas of the Magnet Enclosure. These include the 2 Blower Box hoses, 2 new large Gradient cables, 2 water hoses for Body Coil, and the remainder of the Magnet Enclosure system interconnect cables. Sites with cable access from underneath the Rear Pedestal should have enough cable access area. Sites using the Rear Pedestal cable access area on rear or one side will need to plan to utilize more than one access opening. See Illustrations 2-3 through 2-5 for Magnet cable access areas.

2-4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE
IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER



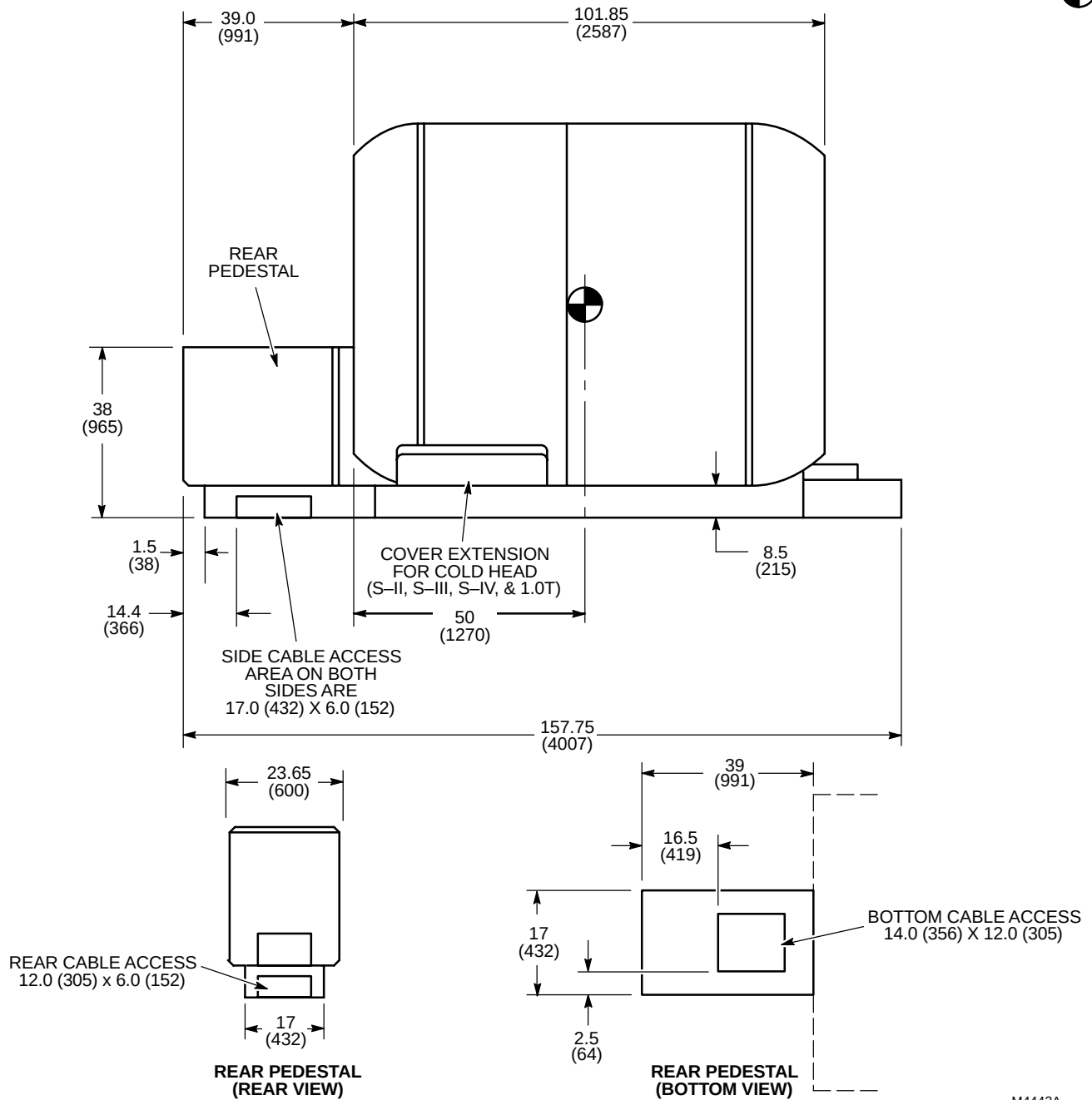
1.0T AND 1.5T Cx (ACTIVE SHIELD) MAGNET CABLE ACCESS

ILLUSTRATION 2-3

2-4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER



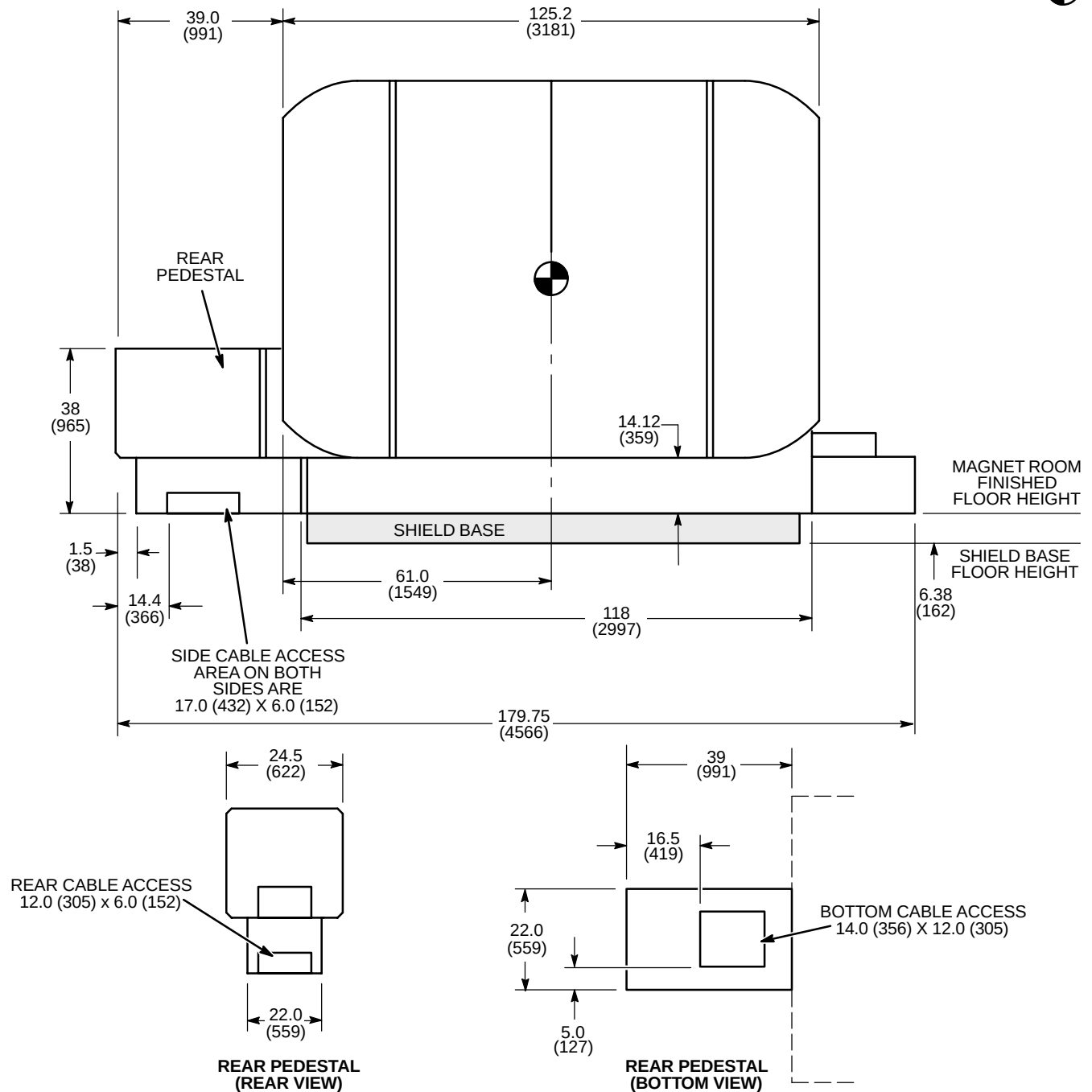
M4442A

1.0T & 1.5T ACTIVE SHIELD, 1.5T UNSHIELDED (S-II, S-III) MAGNET CABLE ACCESS
ILLUSTRATION 2-4

2-4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER



1.5T MAGNET WITH MAGNISHIELD OPTION CABLE ACCESS

ILLUSTRATION 2-5

2-5 SPECIAL SITING CONSIDERATIONS

The following system components have special siting concerns which need to be considered.

2-5-1 Power Distribution Unit (PDU)

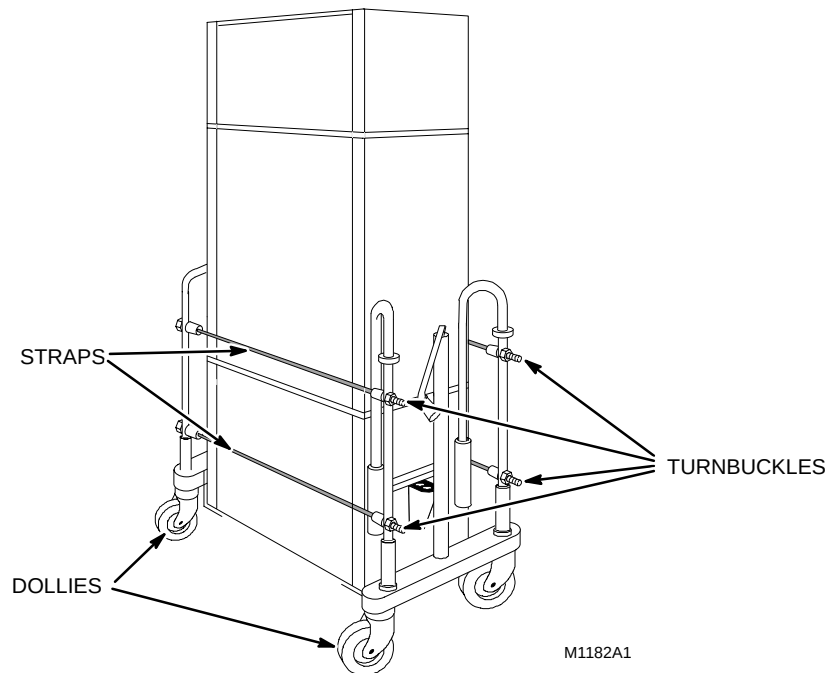
Sites that are upgrading to ACGD require the Full-size PDU, Compact PDU, or RF/PDU Cabinet be de-installed. The Compact PDU and RF/PDU Cabinet have built-in casters for ease of movement. The Full-size PDU does not have casters and weighs approximately 1000 lbs (455 kg). Sites with a Full-size (double bay) PDU must plan for removal of the PDU.

- **Sites located outside the USA:**

The Full-size PDU will have to be removed by riggers or other locally supplied equipment moving personnel/equipment.

- **Sites located in the USA:**

Can contact Techtrans at telephone 800-851-9929 to order a "MR PDU Double Bay Cabinet Dolly Set". The PDU can be moved with two shipping dollies placed one on each side of the PDU and clamped in place using straps with turnbuckles as shown in Illustration 2-6.



FULL-SIZE PDU SHIPPING DOLLIES

ILLUSTRATION 2-6

2-5-2 Blower Box (MG6)

The Blower Box (MG6) provides cooling air for the RF/Gradient Body Coil and the Patient Comfort Module in the Magnet Enclosure. The Blower Box will be mounted within the RF Shielded Room and connect to the Gradient Coil and the Patient Comfort Module by 6.5 inch (165.1 mm) OD and 4.5 inch (114.3 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or mounted to a raised support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room.** The flexible vinyl air ducts route from the Blower Box through the Magnet Enclosure Rear Pedestal cable access and connect to the Gradient Coil. Refer to Illustrations 2-26 for Blower Box and mounting dimensions.

Note

Blower Box mounting requires customer supplied hardware (ie. lag bolts, screws, etc.) appropriate for the surface on which the box will be mounted.

2-5-3 Water Chiller (WC1) For Gradient Coil Cooling

Fixed Site System

A dedicated, closed loop, chilled water cooling system for the Gradient Coil is for fixed site systems utilizing ACGD Gradient driver. The chilled water cooling system consists of a water-to-air Water Chiller (WC1), two flexible vinyl hoses, and de-ionized water and anti-freeze (provided with the coil). See Illustration 2-25 for Water Chiller (WC1) dimension information. The two flexible hoses (supply and return) are to be routed from the Water Chiller (WC1) through waveguides in the Penetration Panel (PP1), through the Rear Pedestal (MG3), and connect to the rear of the Gradient Coil with supplied adjustable compression clamps. The Water Chiller will be powered from the PDU via a supplied power cord. Refer to Section 6, INTERCONNECT DATA, for additional cable information.

Note

The Fixed Site Water Chiller (WC1) for Gradient Coil Cooling must not be located below the Magnet Room floor level. The Water Chiller reservoir tank has a low pressure vent plug which relieves pressure from thermal expansion of water and activates when tank pressure reaches 3-5 psi (0.2-0.3 bar). The vent plug could allow fluid to leak if located below the MR system level. The reservoir tank vented plug can not be replaced with a non-vented type, damage to the tank may occur. The Water Chiller may be located one floor level above the MR system.

Mobile System (True Mobile, Transportable, Relocatables)

The Gradient Coil water cooling will be provided by the Mobile system Van chilled water system. Therefore, the Water Chiller (WC1) as shown in Illustration 2-25 is not used for Mobile configurations.

2-5-4 SPT Phantom Set Shipping/Storage Cabinet

System Performance Test provides the customer and GE Service with a means to quickly verify whether critical parameters affecting image quality are within specifications. The test uses a set of phantoms and a nesting plate for proper positioning of the phantoms on the Patient Table. The phantom set and nesting plate are provided in a cabinet which protects the pieces during shipment and storage at site. The cabinet is not magnetic therefore it can be stored inside the Magnet Room if so desired and moved to the Patient Table for ease of positioning the phantoms. See Illustration 2-38 for dimensions information.

2-5-5 Spectroscopy for 1.5T Magnet Only (Optional)

Upgrade hardware configuration is dependent on the current system cabinets as defined below. Refer to Section 6-7, SPECTROSCOPY INTERCONNECTIONS – 1.5T SYSTEM OPTION, cable length information and other cable data.

Multi-Nuclear Spectroscopy (MNS) For Systems With ACGD/PDU Cabinet

The MNS option provides the Spectroscopy Acquisition Option which consists of the MNS Cabinet which has mounting space for Broadband RF Amplifier and several module kits to modify the SRF Cabinet, Penetration Panel in the Magnet Room, and system cabling.

Multi-Nuclear Spectroscopy (MNS) For Systems With SGD Gradient Cabinet & RF/PDU Cabinet

The MNS option provides the Spectroscopy Acquisition Option which consists of the MNS Cabinet (contains Broadband RF Amplifier) and several module kits to modify the RF/PDU Cabinet, Penetration Panel in the Magnet Room, and system cabling.

Multi-Nuclear Spectroscopy (MNS) For Systems With SGD Gradient Cabinet & RF/Penetration Cabinet

The MNS option provides the Spectroscopy Acquisition Option which consists of the Broadband RF Amplifier (to be installed in the RF/Penetration Cabinet) and several module kits to modify the RF/PDU Cabinet, Penetration Panel in the Magnet Room, and system cabling.

2-5-6 Magnet Monitor

Refer to *Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 2-7-8 Magnet Monitor* for LCC Magnet Monitor and Magnet Monitor UPS option information.

2-5-7 System Monitoring & Support Connectivity Requirements

One of the system monitoring and support connectivity configurations listed in Table 2-4 must be provided for system installation and serviceability purposes. The network connection (if selected configuration) and telephone lines are to be provided and paid for by the customer.

TABLE 2-4
SYSTEM MONITORING & SUPPORT CONNECTIVITY REQUIREMENTS

CONFIGURATION	PHONE LINE	USE/LOCATION
Network connection & telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	Network connection with a static IP address See Note 1	Access located near the Operator Workspace (OW) in the Control Room.
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring.
Multiple telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Operator Workspace (OW) in the Control Room. See Note 2 & 3
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. See Note 2 & 3
Note 1 For North America & South America: One Internet accessible Virtual Private Network Connection with a static IP Address is required. For Europe: One ISDN Network Connection with a static IP Address is required. 2 A dedicated direct-distance-dialing <u>voice-grade</u> telephone line can be shared for Operator Workspace (OW) and Magnet Monitor (MSM) requirement through the use of a multiplexer box. The following multiplexer boxes are available for customer purchase. 46-328475P1 4 Line Phone Multiplexer box; 115 VAC input power 46-328475P3 4 Line Phone Multiplexer box; 220 VAC input power If the customer chooses not to purchase the multiplexer box then the customer must provide an additional line for each requirement as stated in this table. 3 If a Multiplexer Box is used then the Magnet Monitor MUST be Channel 1 to allow for call out after a power outage.		

2-6 ARCHITECTURAL REMINDERS

- The operator seated at the Operator Workspace must have an unobstructed view of the patient on the transport table when table is docked to the magnet.
- It is recommended that the Magnet Room viewing window be of fine mesh screening material (as opposed to a "honeycomb-type pattern") for better visibility of the patient from the Operator Workspace.
- Customer provided and paid for telephone lines must be supplied for system installation and serviceability purposes per **Section 2-5-7 System Monitoring & Support Connectivity Requirements**.
- Recommend protecting floors while moving heavy pieces of equipment (e.g. ACGD/PDU Cabinet, BRM Body Coil Assembly).

2-7 FLOOR LOADING AND WEIGHTS

This section contains loading considerations for Signa Horizon upgrade components. Listed in Table 2-5 are the weights, floor loading, and normal mounting methods for components.

TABLE 2-5
SIGNA HORIZON LX FLOOR LOADING

COMPONENT	NET WT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
Cx Magnet, Enclosure, and Coils	See Note 1	87.5 x 83.4 x 91.9 (2223 x 2119 x 2333)	Not changed by Horizon upgrade.	Not changed by Horizon upgrades.
Oxford & S-Series Magnet, Enclosure, and Coils	See Note 1	90.62 x 157.75 x 96.36 (2302 x 4007 x 2448)	Not changed by Horizon upgrade.	Not changed by Horizon upgrades.
Magnet, Magnishield, Enclosure, and Coils	See Note 1	107 x 179.75 x 96.36 (2718 x 4566 x 2448)	Not changed by Horizon upgrade.	Not changed by Horizon upgrades.
Patient Table	No change to existing Lightweight Patient Transport.			
Blower Box	47 (21)	16 x 16 x 22 (406 x 406 x 559)	Square base 15 x 15 (381 x 381)	Anchor to floor, shelf, or wall.
Power Distribution Unit	Existing PDU is modified internally, floor loading not impacted.			
Water Chiller for Gradient Coil	374 (170)	21.65 x 27.95 x 37 (550 x 710 x 940)	Rectangular base 21.65 x 27.95 (550 x 710). Four casters 3 (76) dia. x .75 (19) wide.	Set on floor and rest on casters.
SRF Cabinet	525 (239)	24 x 40 x 70.7 (610 x 1016 x 1793)	Rectangular base 24 x 30 (610 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5T MR/i with EXCITE Technology System Cabinet	468 (213)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
1.5T & 1.0T Signa MR/i or 1.5T Signa CV/i System Cabinet	494 (225)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
ACGD/PDU Cabinet	1844 (838)	23.5 x 37.25 x 85.6 (597 x 927 x 2174) See Note 3	Rectangular base 33.5 x 22 (851 x 559). Four casters each 3.0 (76.2) dia. x 1.82 (46.2) wide.	Set on floor on casters.
Operator Workspace Table with LCD Color Monitor (See Note 4)	175 (80)	54 x 43 x 52 (1372 x 1092 x 1321)	See Illustration 2-33..	Set on floor: Table on leveling pads & Cabinet rest on casters. Anchor Table to floor per Section 2-7-1.
Operator Workspace Cabinet (See Note 4)	192 (87)	18.5 x 29 x 26 (470 x 737 x 660)	See Illustration 2-34.	Set on floor and rest on casters.
(Continued)				

TABLE 2-5 (Continued)
SIGNA HORIZON LX FLOOR LOADING

COMPONENT	NET WT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
MNS Cabinet Option	430 (195)	23.56 x 42 x 70.7 (598 x 1067 x 1797)	Rectangular base 33.5 x 22 (851 x 559).	Casters for location. Set on floor on four leveling pads.
Note 1 Weight of magnet with enclosure, shim coil, RF/Gradient Body Coil Assembly, and cryogenics is: –1.0T Cx Active Shield11,856 lbs (5378 kg) –1.0T (S–X, S–XC) Active Shield20,700 lbs (9409 kg) –1.5T Cx Active Shield12,606 lbs (5719 kg) –1.5T (S–IV, S–V) Active Shield20,600 lbs (9364 kg) –1.5T (S–III) Unshielded18,420 lbs (8373 kg) –1.5T (S–II) Unshielded22,020 lbs (10,010 kg) –1.5T (S–I) Unshielded15,625 lbs (7102 kg) –1.5T (S–III)magnet and Magnishield82,420 lbs (37,464 kg) 2 Consult a structural engineer on method of calculating proper weight/unit area for floor loading. 3 ACGD/PDU Cabinet height with Fan Module removed is 74.8 inches (1900 mm). 4 The Operator Workspace Cabinet minimum area location is under the Workspace Table, see in Illustration 2–33. An alternate location is possible with the Operator Workspace Cabinet located against the right side of the Workspace Table.				

2-7-1 Operator Workspace Mounting Requirements For Signa LX

The Operator Workspace Table sets on the floor with the Workspace Cabinet positioned under the Table towards the right side, see Illustration 2-33. Note, the Workspace Table must be bolted to the floor for safe use of the table and equipment.

2-8 COMPONENT DIMENSIONS

To assist in completing your room layout, refer to Table 2-6 for list of component Illustrations and the upgrade catalog numbers which provide the new equipment. Included are items which may impact site layout or need to be taken into consideration when determining delivery route. Modifications to existing cabinets are not included, catalogs may list the same new equipment but differ due to other modification hardware.

TABLE 2-6
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M1000MN M1000MP M1000MR M1000MS M1000MW M1000NH M1000NJ M1000NK M1000NT	OPERATOR WORKSPACE (OW1)	2-33
M1033K M1033L	OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER	2-36
M1033MD	MNS CABINET (MNS) – 1.5T OPTION	2-39
M1060JW	LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)	2-24
M1060LC	1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN	2-22
	MAGNET MONITOR (MSM1)	2-23
M1085BG	UPGRADE RF/GRADIENT (BRM OR CRM) COIL & CRADLE ON CART	2-2
M1085KB M1085KC	WATER CHILLER (WC1) FOR GRADIENT COIL COOLING WATER	2-25
M1090TL	PATIENT TRANSPORT TABLE (PT1)	2-21
M1090JP	ENCLOSURE REAR END BELL ASSEMBLY – STANDARD MAGNET	2-12
	ENCLOSURE FRONT COVER – MAGNISHIELD MAGNET	2-18
	ENCLOSURE FRONT END BELL ASSEMBLY – MAGNISHIELD MAGNET	2-19
	ENCLOSURE REAR COVER – MAGNISHIELD MAGNET	2-20
	BLOWER BOX (MG6)	2-26
	REAR PEDESTAL	See Note 1
M1090JR	ENCLOSURE FRONT COVER – STANDARD	2-7
	ENCLOSURE FRONT END BELL ASSEMBLY – STANDARD MAGNET	2-8
	ENCLOSURE REAR COVER – STANDARD MAGNET	2-11
	ENCLOSURE REAR END BELL ASSEMBLY – STANDARD	2-12
	BLOWER BOX (MG6)	2-26
	REAR PEDESTAL	See Note 1
M1090JS M1090JT	BRIDGE – STANDARD	2-16
M1090SC	BRIDGE – MAGNISHIELD	2-17
M1090TL	PATIENT TABLE (PT1)	2-21
Note 1 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2-3, MINIMUM DOOR/HALLWAY SIZES, for dimension information.		
(Continued)		

2-8 COMPONENT DIMENSIONS (Continued)

TABLE 2-6 (Continued)
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M1191JF	ENCLOSURE FRONT COVER – STANDARD MAGNET	2-7
	ENCLOSURE FRONT END BELL ASSEMBLY – STANDARD MAGNET	2-8
	ENCLOSURE REAR END BELL ASSEMBLY – STANDARD MAGNET	2-12
	BLOWER BOX (MG6)	2-26
	REAR PEDESTAL	See Note 1
	ENCLOSURE REAR COVER – MOBILE RF TUBE EXTENSION	2-15
M1860LC	1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN	2-22
	MAGNET MONITOR (MSM1)	2-23
M1885BG	UPGRADE RF/GRADIENT BRM COIL & CRADLE ON CART	2-2
M3090AA	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CONTROL CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	PENETRATION PANEL (PP1)	2-31
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090AD	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	PENETRATION PANEL (PP1)	2-31
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3090AH	ACGD/PDU CABINET (MR3)	2-27
M3090AH	SYSTEM CABINET (MR2)	2-29
M3090AH	SRF CABINET (MR1)	2-30
M3090AH	OPERATOR WORKSPACE (OW1)	2-33
M3090AH	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3090BA	ACGD/PDU CABINET (MR3)	2-27
M3090BA	SRF CABINET (MR1)	2-30
M3090BB	ACGD/PDU CABINET (MR3)	2-27
M3090BB	SRF CABINET (MR1)	2-30
M3090BC	ACGD/PDU CABINET (MR3)	2-27
M3090BC	SRF CABINET (MR1)	2-30
M3090AE	ACGD/PDU CABINET (MR3)	2-27
M3090AE	SYSTEM CABINET (MR2)	2-29
M3090AE	SRF CABINET (MR1)	2-30
M3090AE	PENETRATION PANEL (PP1)	2-31
M3090AE	OPERATOR WORKSPACE (OW1)	2-33
M3090AE	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
Note 1 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2-3, MINIMUM DOOR/HALLWAY SIZES, for dimension information.		
(Continued)		

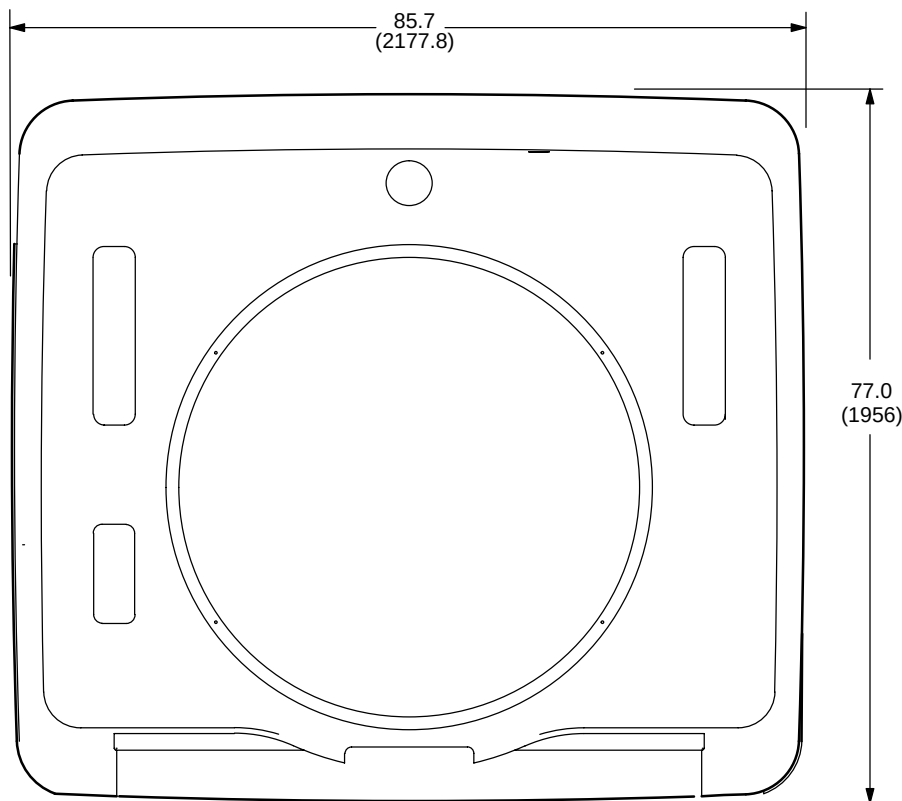
2-8 COMPONENT DIMENSIONS (Continued)

TABLE 2-6 (Continued)
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

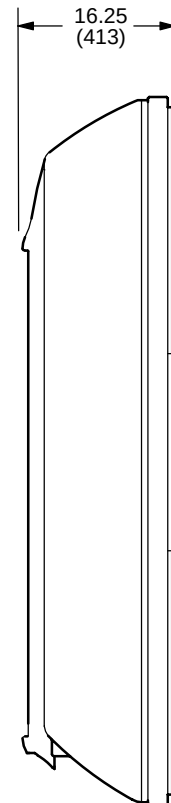
UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M3090AG	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3090CA	PENETRATION PANEL COVER	2-32
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CB	PENETRATION PANEL COVER	2-32
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CC	PENETRATION PANEL COVER	2-32
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CH	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CJ	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090CK M3090CM	PENETRATION PANEL COVER	2-32
	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-38
M3090DA M3090DB M3090DC	MNS CABINET (MNS) – 1.5T OPTION	2-39
M3090EC	SYSTEM CABINET (MR2) FOR 1.5T SIGNA MR/i WITH EXCITE TECHNOLOGY	2-28
M3090ED M3090EE	ACGD/PDU CABINET (MR3)	2-27
	SRF CABINET (MR1)	2-30
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3890AB	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	PENETRATION PANEL (PP1)	2-31
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
M3890AD	ACGD/PDU CABINET (MR3)	2-27
	SYSTEM CABINET (MR2)	2-29
	SRF CABINET (MR1)	2-30
	OPERATOR WORKSPACE (OW1)	2-33
	OPERATOR WORKSPACE CABINET (OW1 A2)	2-34
Note 1 Catalog provides a kit to modify existing system equipment to the configuration shown in the illustration. 2 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2-3, MINIMUM DOOR/HALLWAY SIZES, for dimension information.		

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 125 lbs (57 kg)



FRONT VIEW

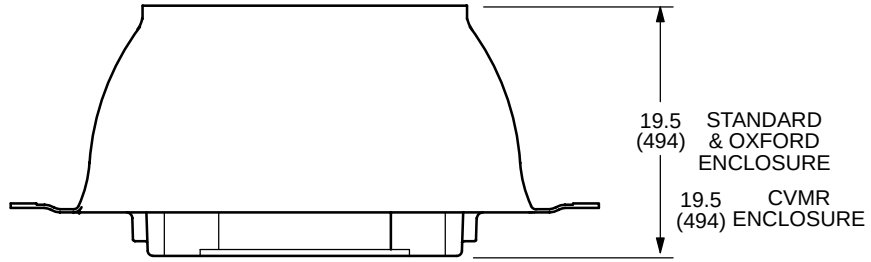


SIDE VIEW

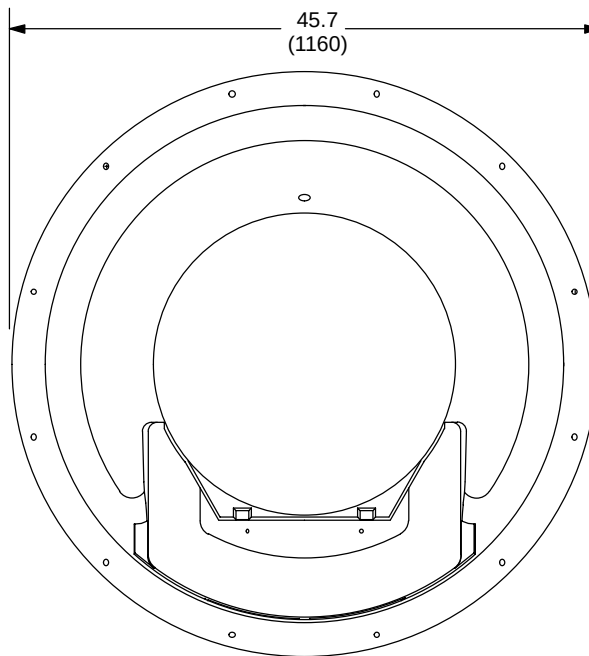
ENCLOSURE FRONT COVER – STANDARD
ILLUSTRATION 2-7

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 42 lbs (19 kg)
- STANDARD ENCLOSURE END BELL SHOWN.
OXFORD ENCLOSURE END BELL HAS SAME
DIMENSIONS WITH MINOR FIT AND
APPEARANCE DIFFERENCES.



TOP VIEW

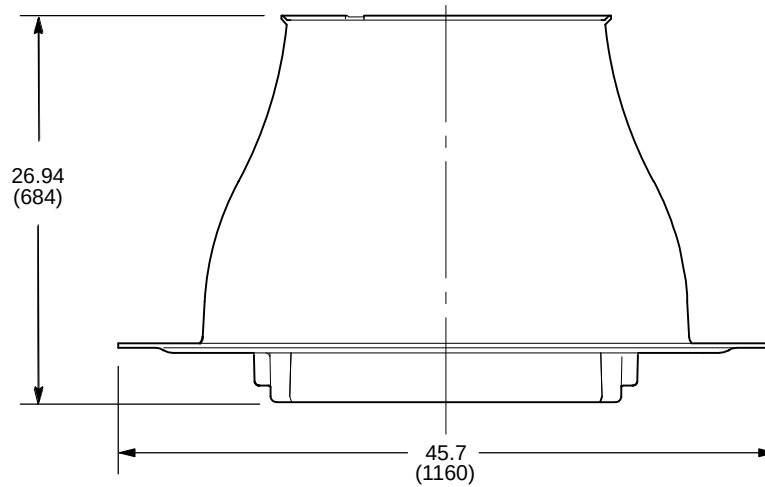


FRONT VIEW

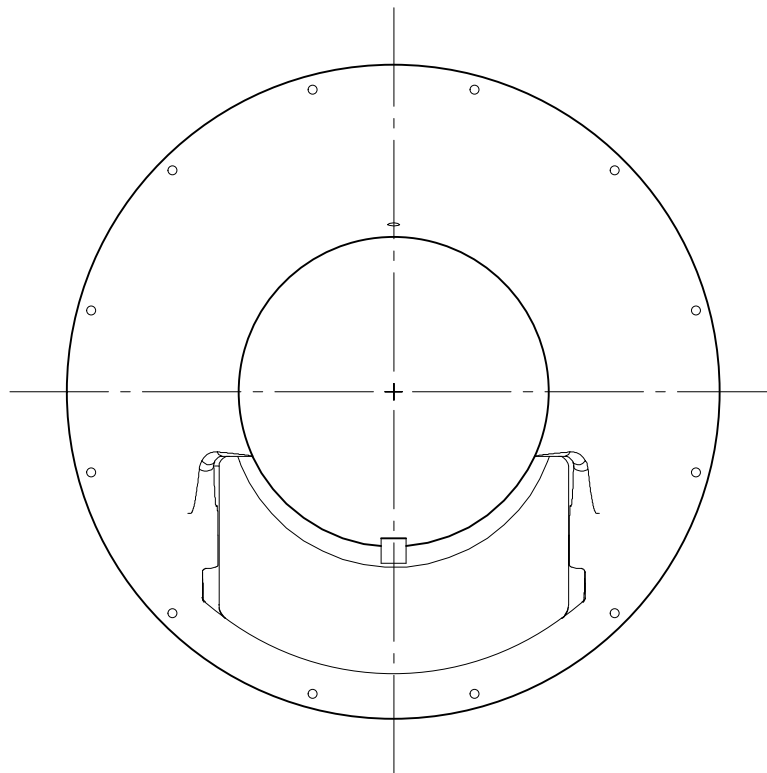
ENCLOSURE FRONT END BELL ASSEMBLY – STANDARD MAGNET
ILLUSTRATION 2–8

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)



TOP VIEW

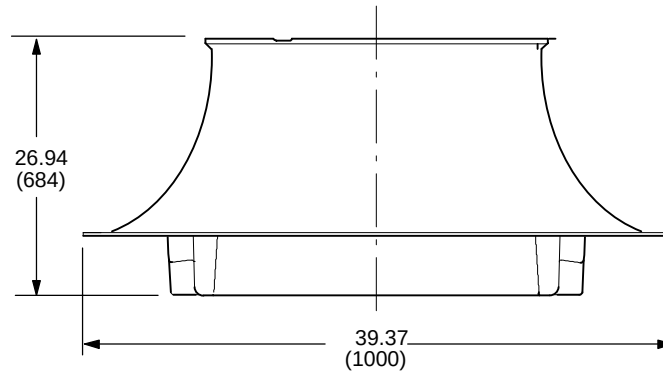


FRONT VIEW

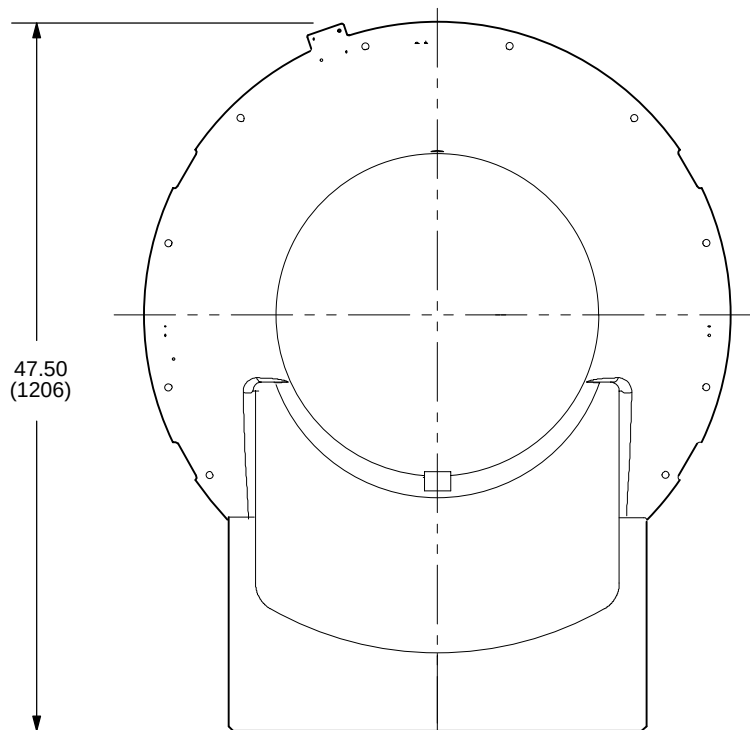
ENCLOSURE FRONT END BELL ASSEMBLY FOR S-II THROUGH S-V MAGNETS – SIGNA CVMR
ILLUSTRATION 2-9

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)



TOP VIEW



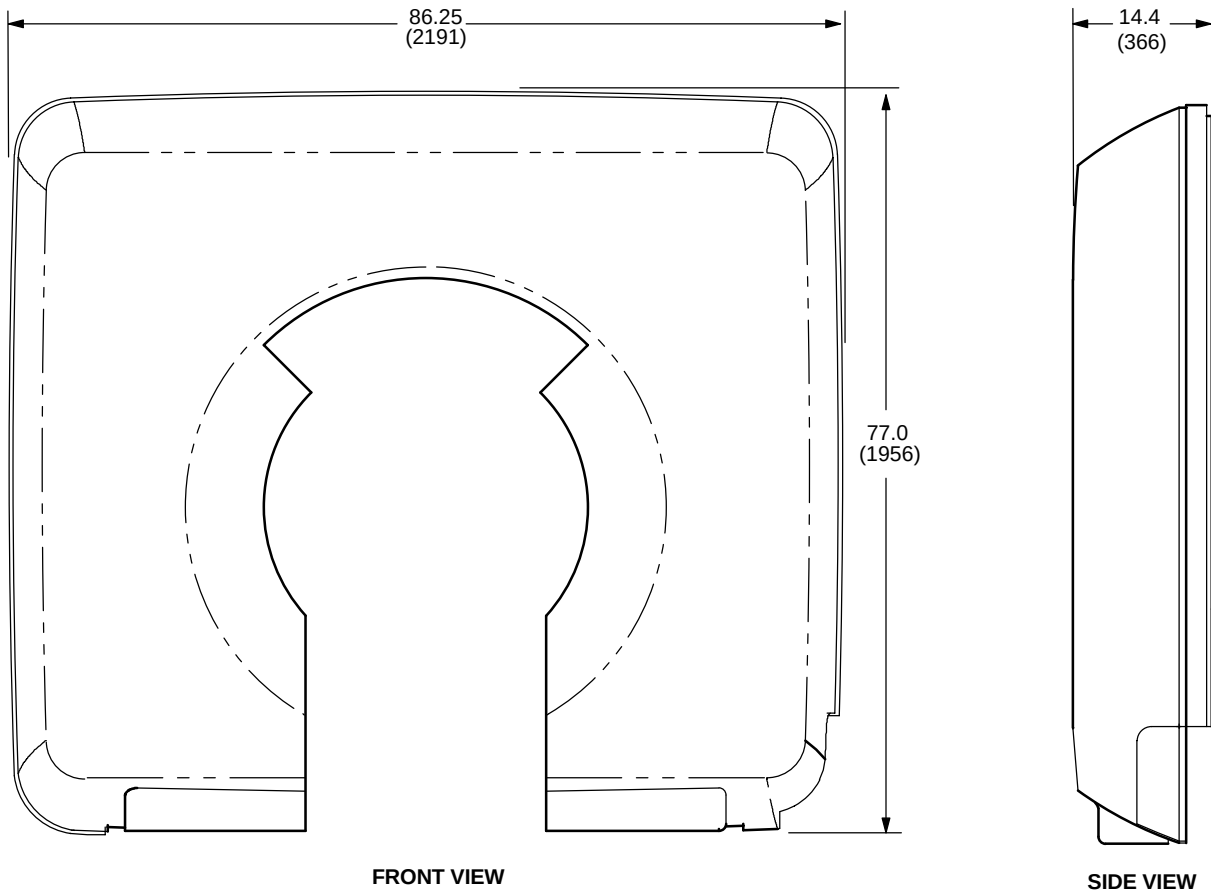
FRONT VIEW

M

ENCLOSURE FRONT END BELL ASSEMBLY FOR Cx & LCC MAGNETS
ILLUSTRATION 2-10

NOTE:

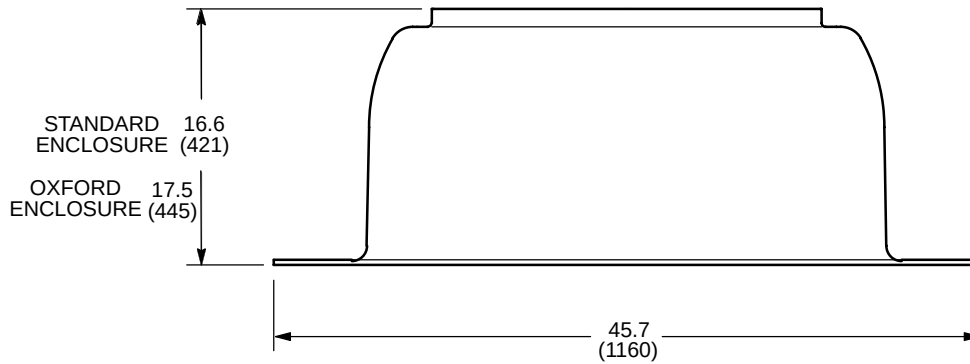
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 140 lbs (64 kg)



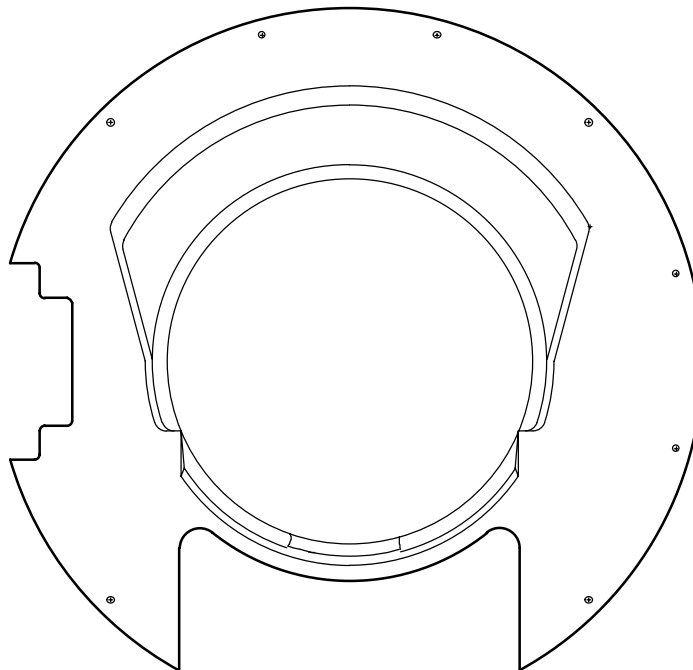
ENCLOSURE REAR COVER – STANDARD
ILLUSTRATION 2-11

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)
- STANDARD ENCLOSURE END BELL SHOWN.
OXFORD ENCLOSURE END BELL HAS MINOR
FIT AND APPEARANCE DIFFERENCES IN
ADDITION TO DIMENSION DIFFERENCE
NOTED.



TOP VIEW

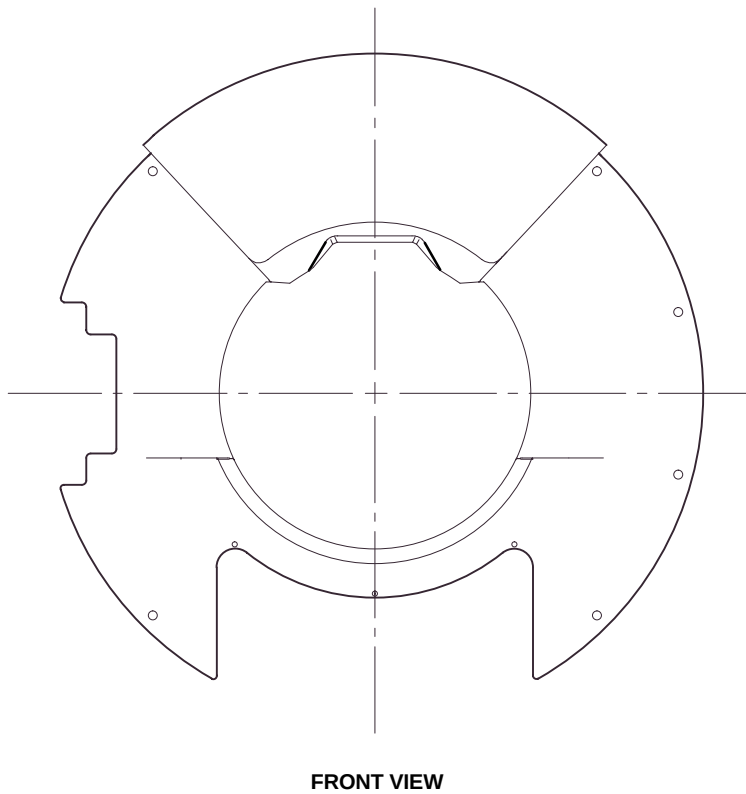
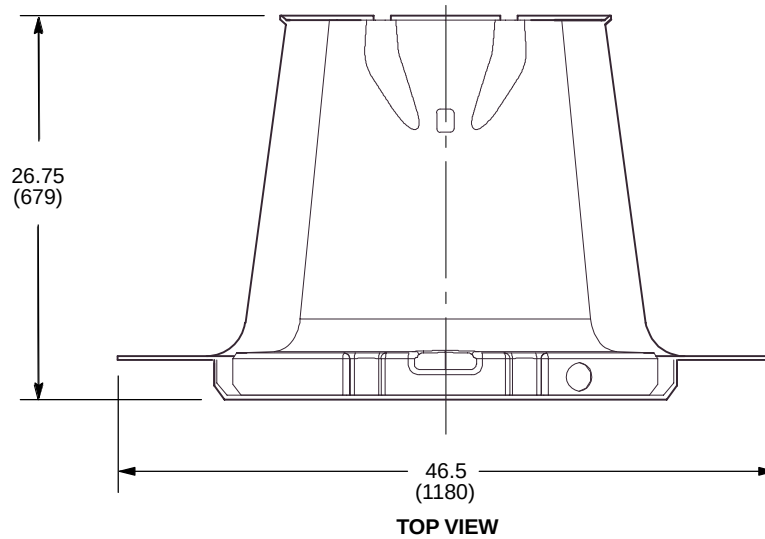


FRONT VIEW

ENCLOSURE REAR END BELL ASSEMBLY – STANDARD MAGNET
ILLUSTRATION 2-12

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)

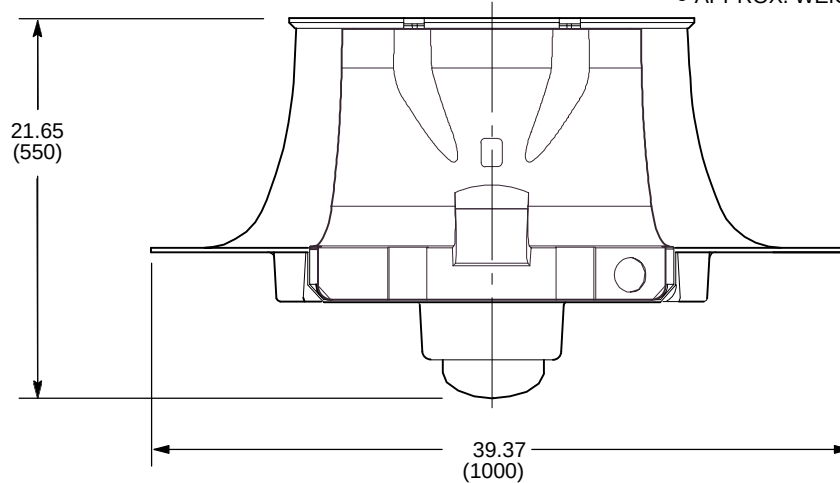


ENCLOSURE REAR END BELL ASSEMBLY FOR S-II THROUGH S-V MAGNETS
ILLUSTRATION 2-13

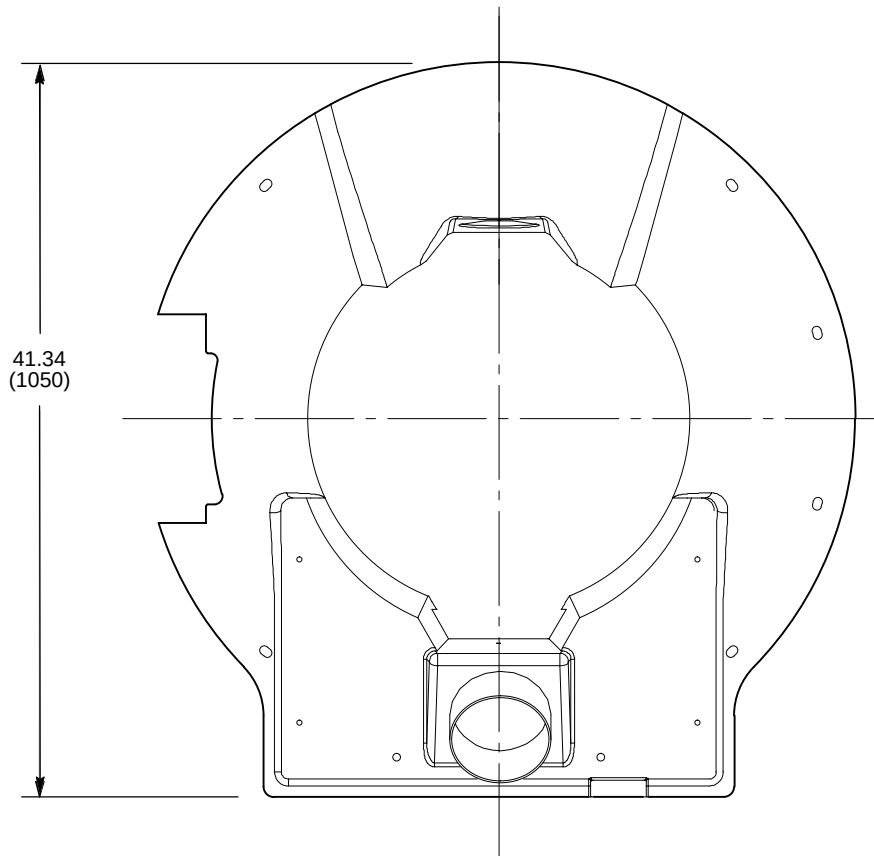
M

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN
MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (21 kg)



TOP VIEW



FRONT VIEW

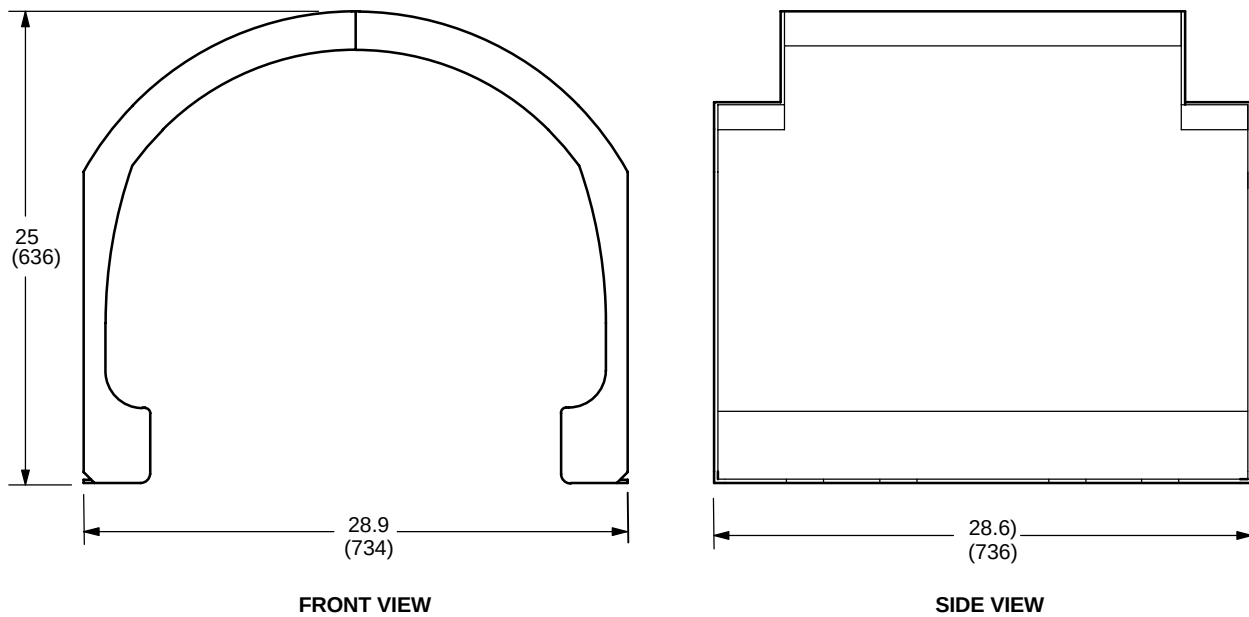
ENCLOSURE REAR END BELL ASSEMBLY FOR Cx & LCC MAGNETS

ILLUSTRATION 2-14

M

NOTE:

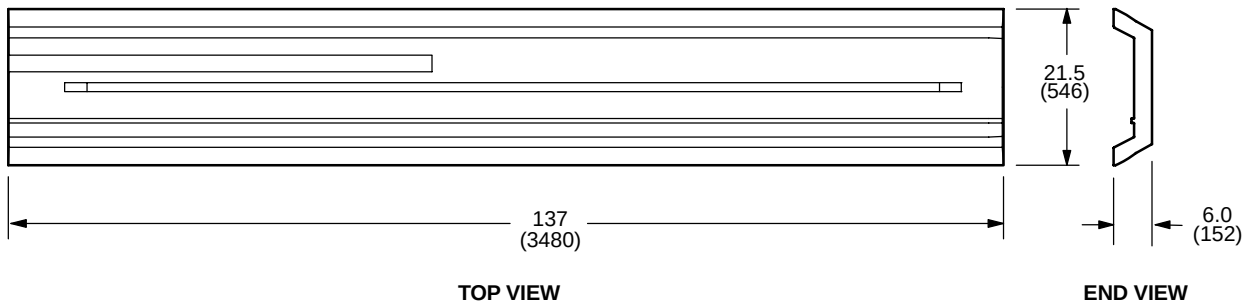
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 25 lbs (11 kg)



ENCLOSURE REAR COVER - MOBILE RF TUBE EXTENSION
ILLUSTRATION 2-15

NOTE:

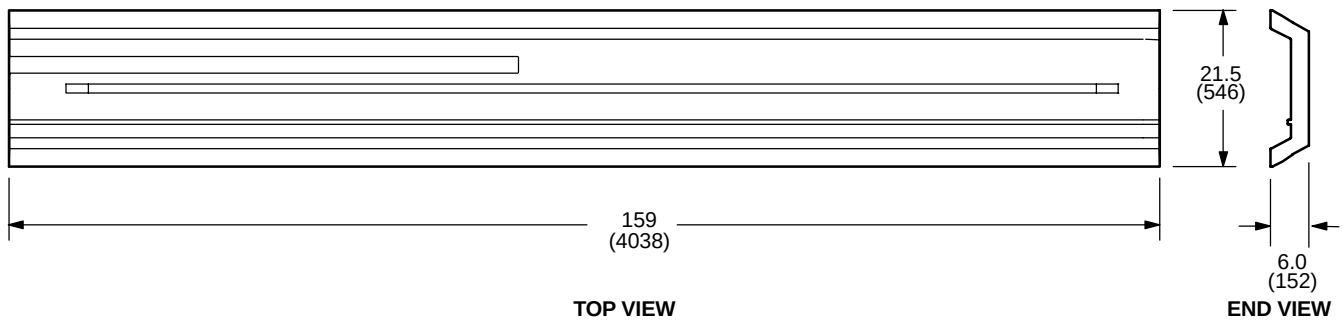
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



BRIDGE – STANDARD ENCLOSURE
ILLUSTRATION 2-16

NOTE:

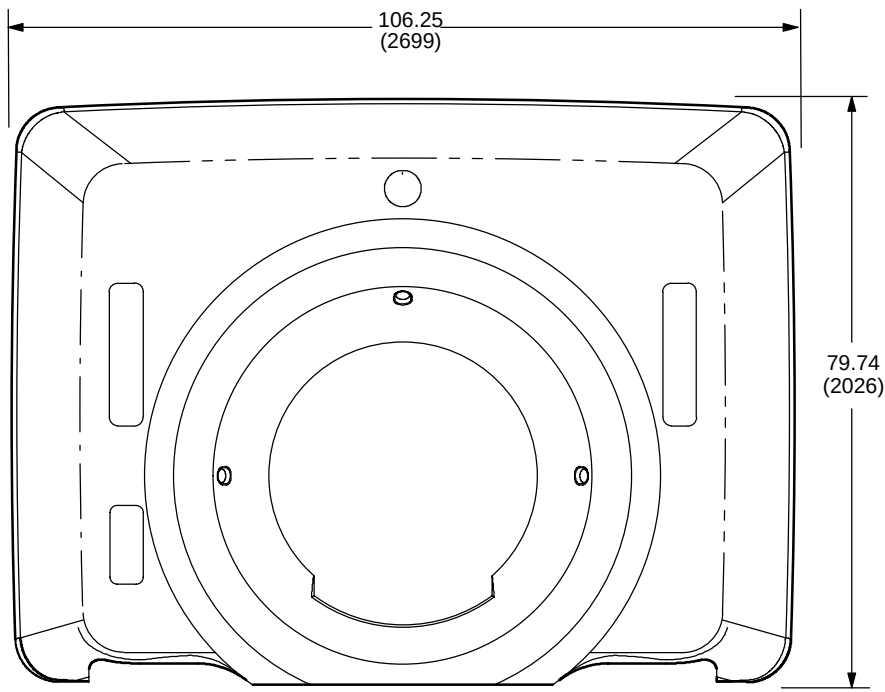
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 90 lbs (39 kg)



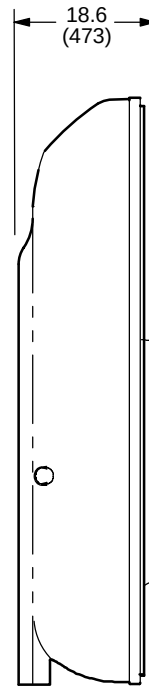
BRIDGE – MAGNISHIELD ENCLOSURE
ILLUSTRATION 2-17

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 170 lbs (77 kg)



FRONT VIEW

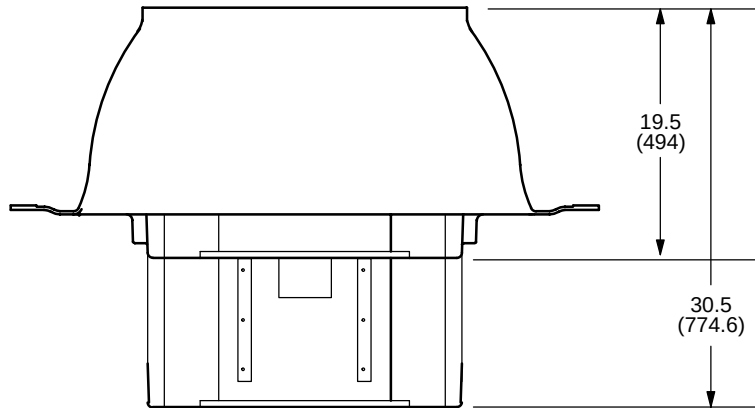


SIDE VIEW

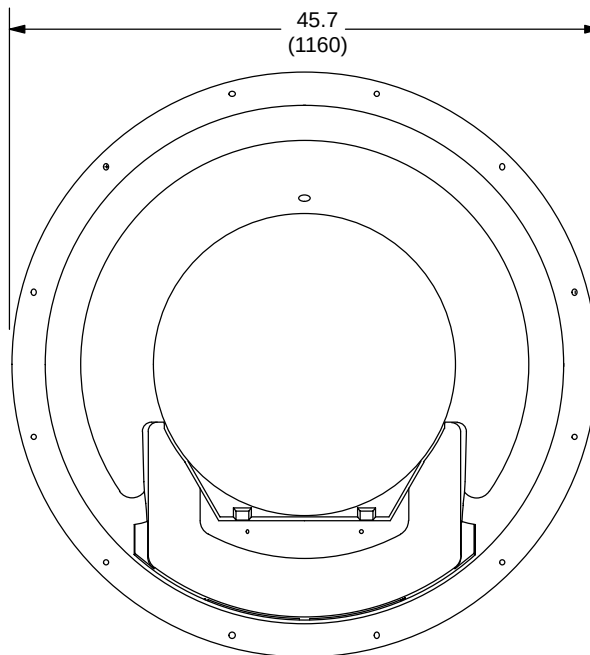
ENCLOSURE FRONT COVER – MAGNISHIELD MAGNET
ILLUSTRATION 2-18

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 55 lbs (22 kg)
- MAGNISHIELD ENCLOSURE FRONT END BELL CONSISTS OF STANDARD ENCLOSURE FRONT END BELL AND MAGNISHIELD END BELL EXTENSION INSTALLED DURING UPGRADE.



TOP VIEW



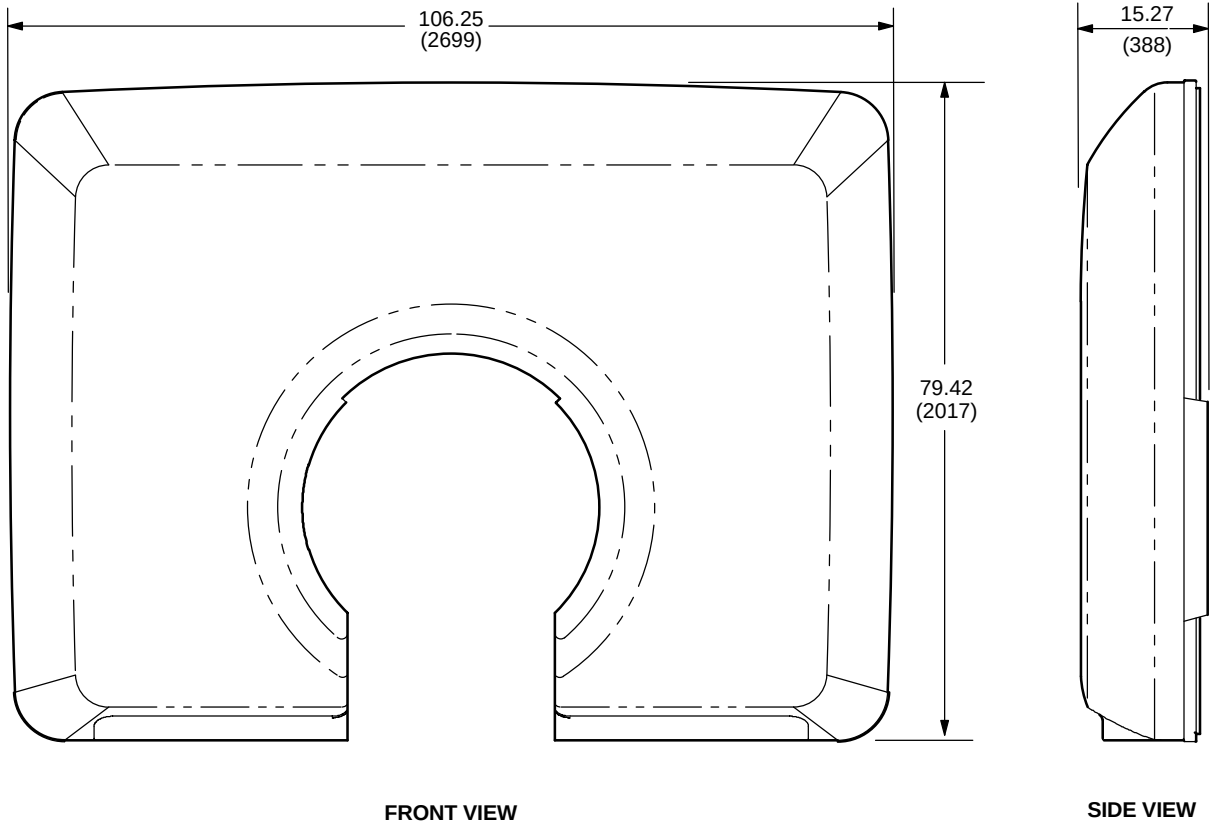
FRONT VIEW

ENCLOSURE FRONT END BELL ASSEMBLY – MAGNISHIELD

ILLUSTRATION 2-19

NOTE:

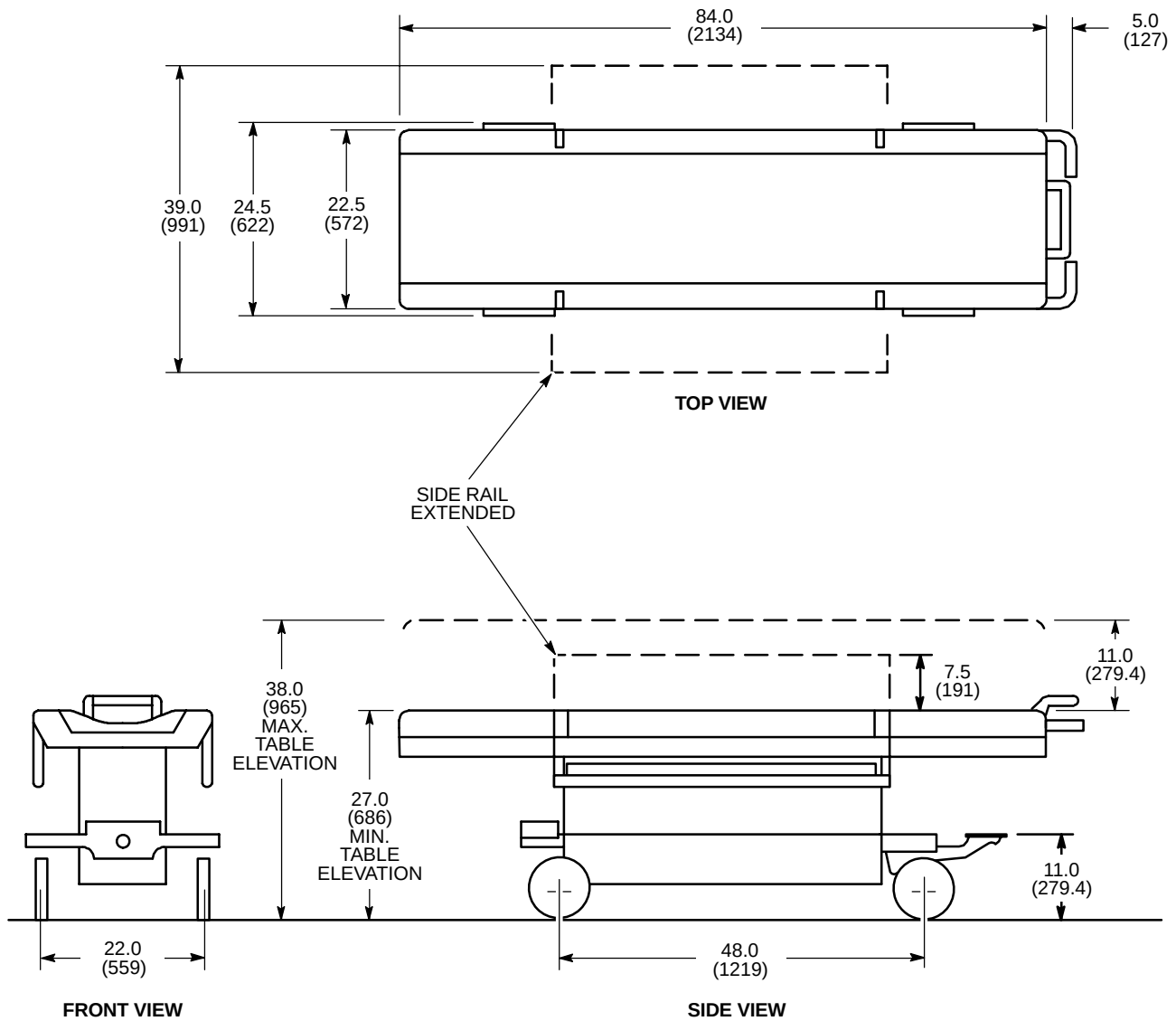
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 170 lbs (77 kg)



ENCLOSURE REAR COVER – MAGNISHIELD MAGNET
ILLUSTRATION 2-20

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- WEIGHT: 280 lbs (127 kg)



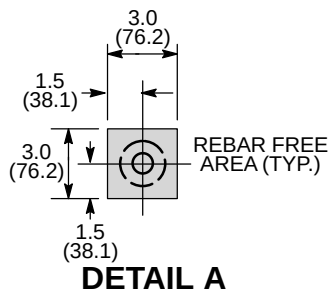
M1081A2

PATIENT TABLE (PT1)
ILLUSTRATION 2-21

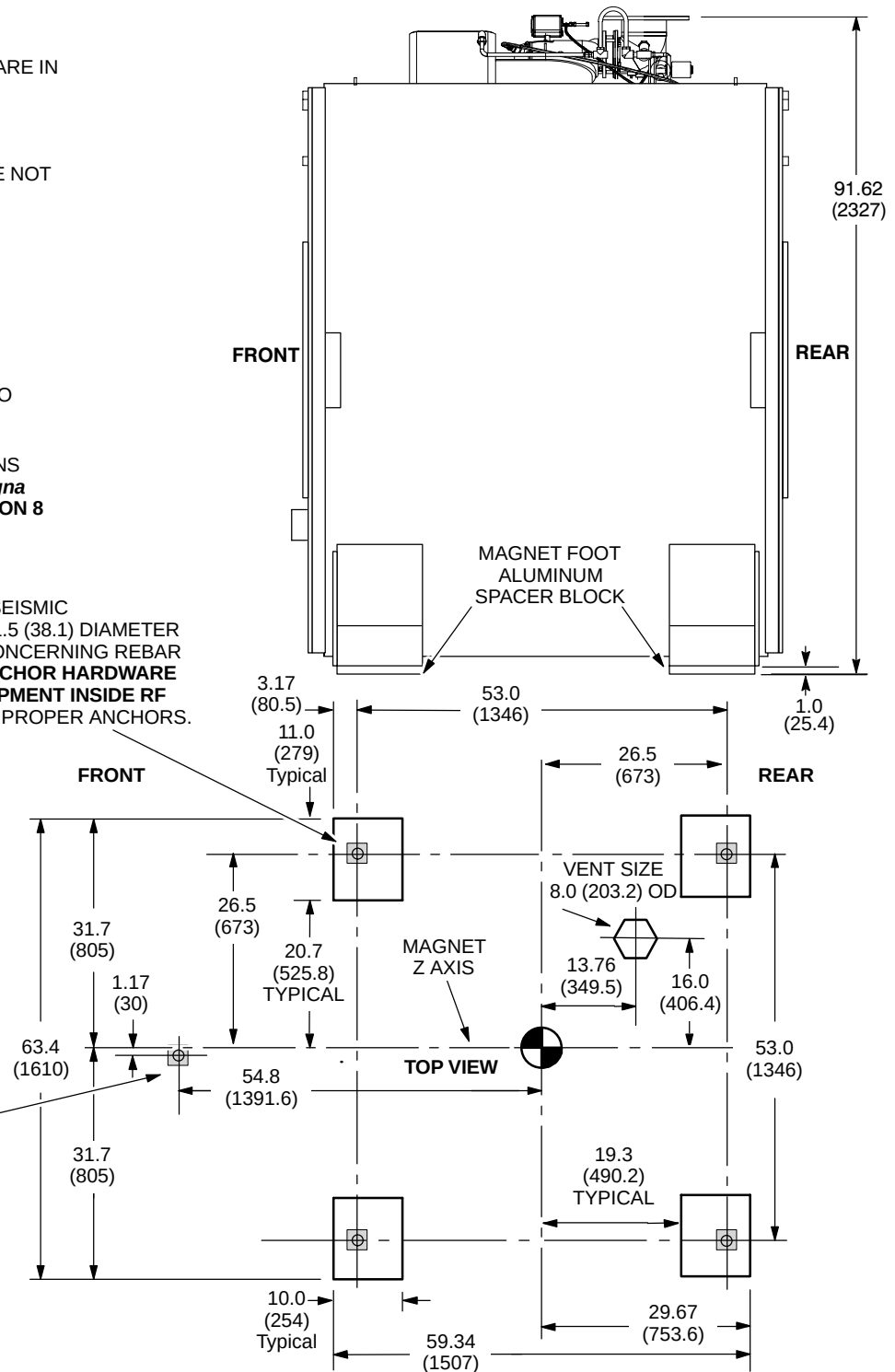
NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- LIFTING/JACKING BEAMS ARE ATTACHED TO SIDES OF MAGNET. REFER TO ILLUSTRATION 8-1.
- ALUMINUM LEVELING PLATES ARE NOT SHOWN.
- APPROX. WEIGHT OF MAGNET, CRYOGENS, COILS, AND ENCLOSURE WITH BRM
1.0T: 11,856 lbs (5378 kg)
1.5T: 12,606 lbs (5719 kg)
FOR SIGNA CV/i WITH CRM
1.5T: 12,806 lbs (5821 kg)
- STEEL REBAR MUST NOT BE POSITIONED IN SHADED AREAS TO PREVENT INTERFERENCE WITH MOUNTING BOLTS.
- FOR MAGNET MOVING DIMENSIONS REFER TO **DIRECTION 2223170 Signa MR/i & CV/i Pre-Installation SECTION 8 SHIPPING AND DELIVERY DATA**

SEISMIC AND NON-SEISMIC
MAGNET MOUNTING HOLES (4) 1.5 (38.1) DIAMETER
SEE NOTE ABOVE & DETAIL A CONCERNING REBAR
REFER TO **SECTION NO TAG ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** TO DETERMINE PROPER ANCHORS.

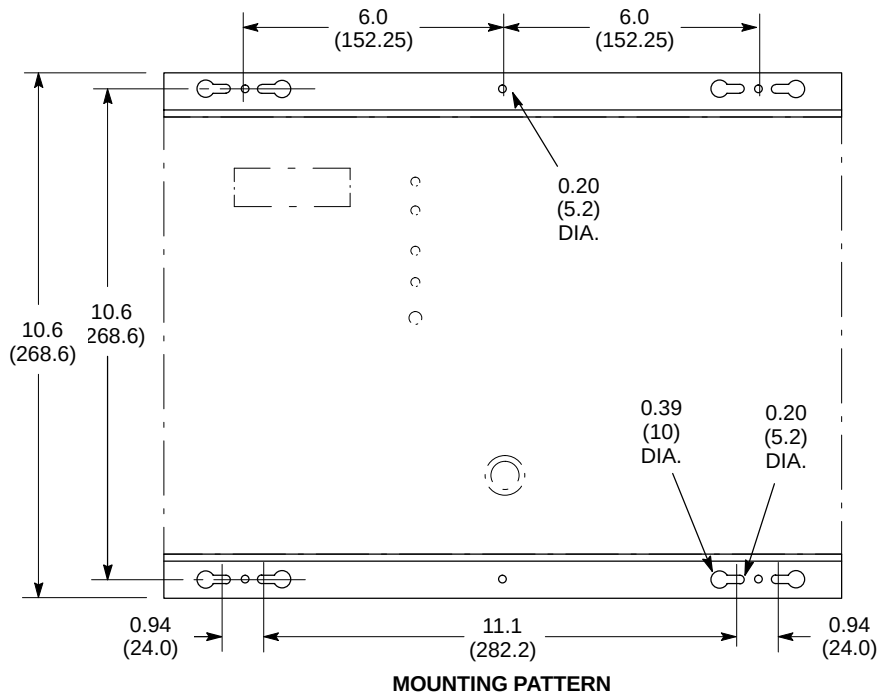


M10 ANCHOR FOR
TABLE DOCK MOUNTING,
LOCATED OFFSET FROM
MAGNET CENTER LINE.
FOR ADDITIONAL
MOUNTING INFORMATION
REFER TO SECTION
NO TAG
**ANCHOR HARDWARE
REQUIREMENTS FOR MR
EQUIPMENT INSIDE RF
SHIELDED ROOM.**



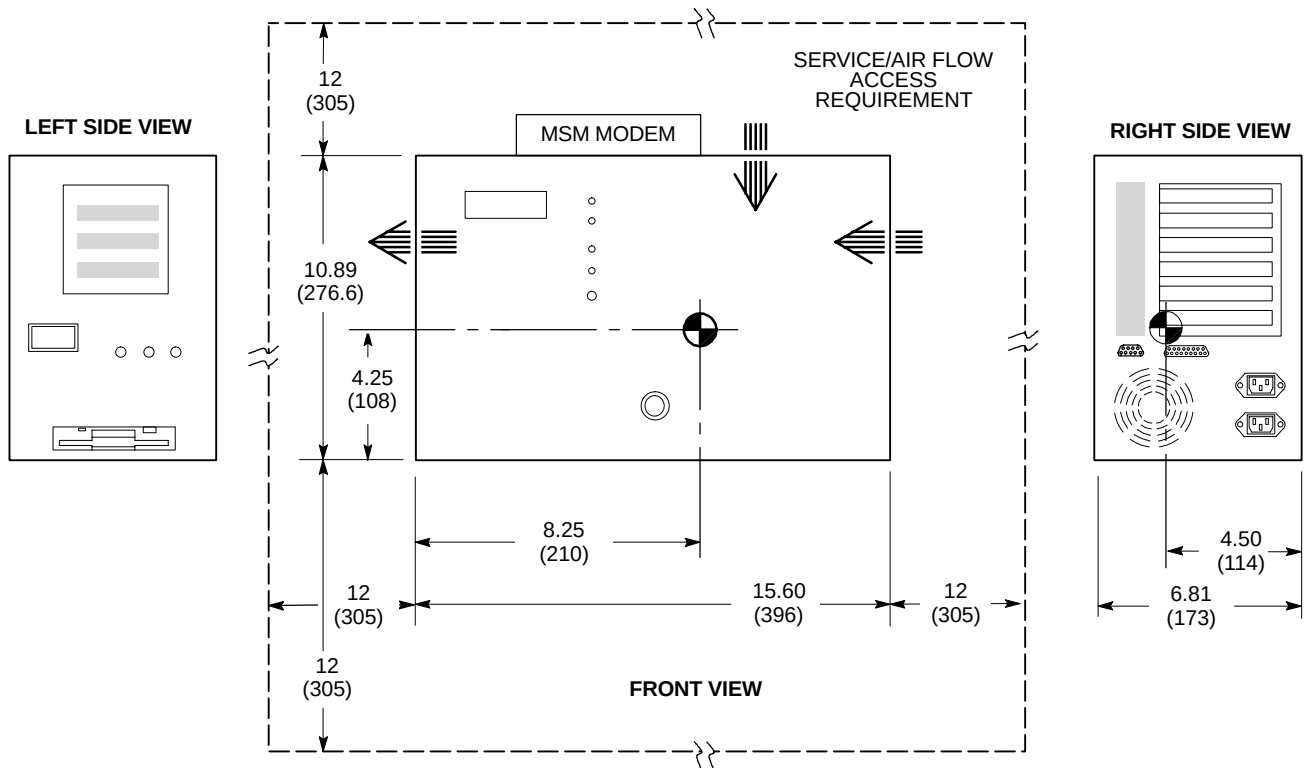
1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN

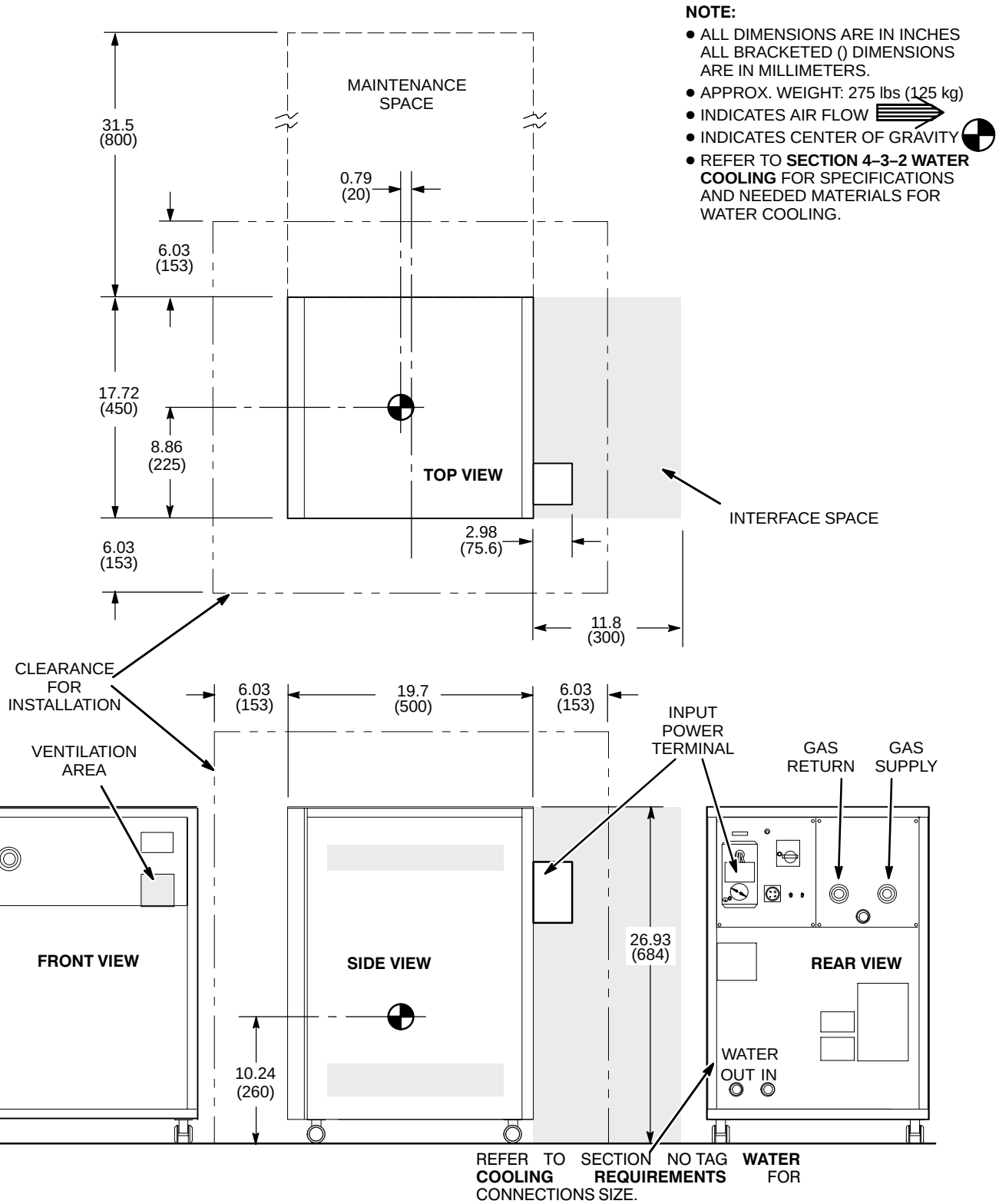
ILLUSTRATION 2-22



NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 22 lbs (10 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY



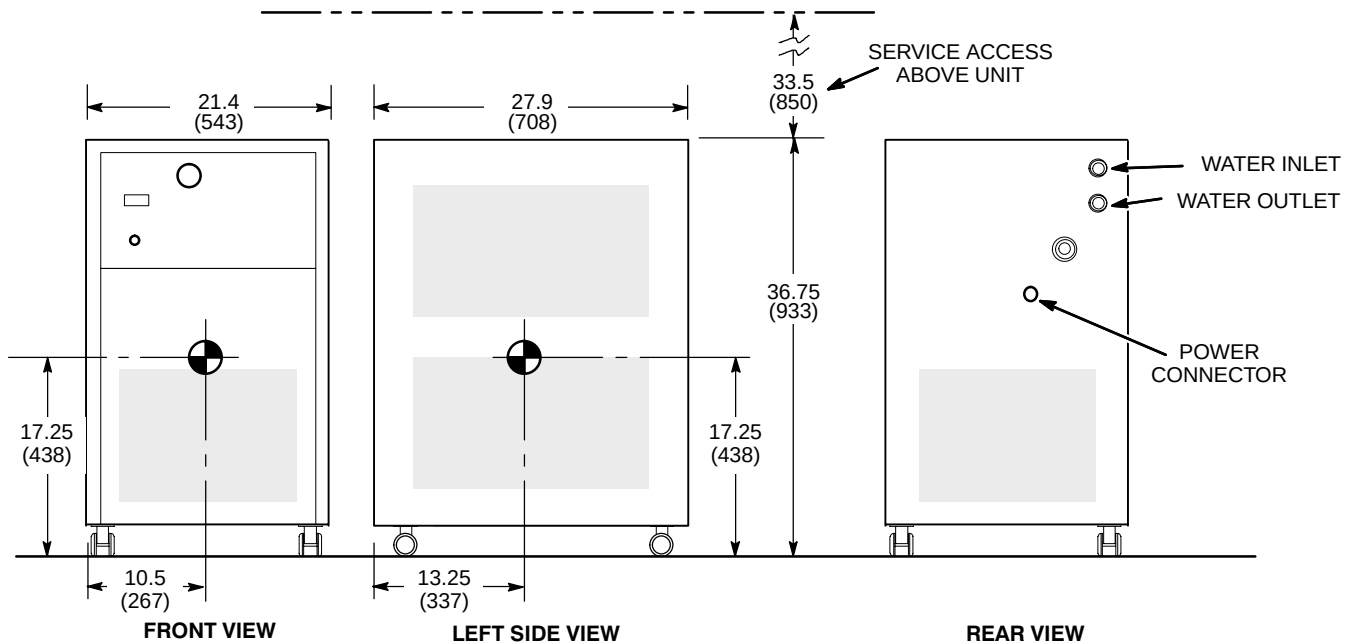
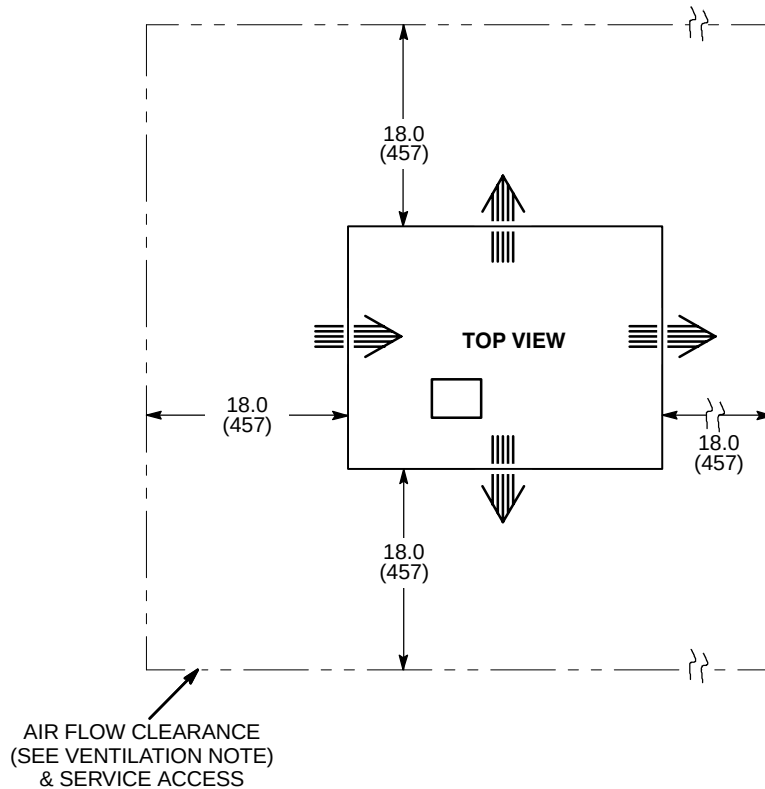


LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)

ILLUSTRATION 2-24

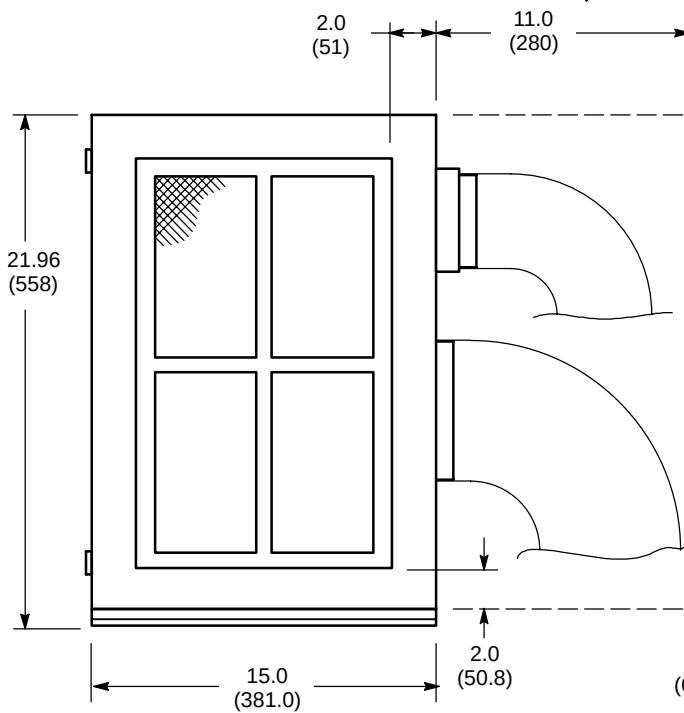
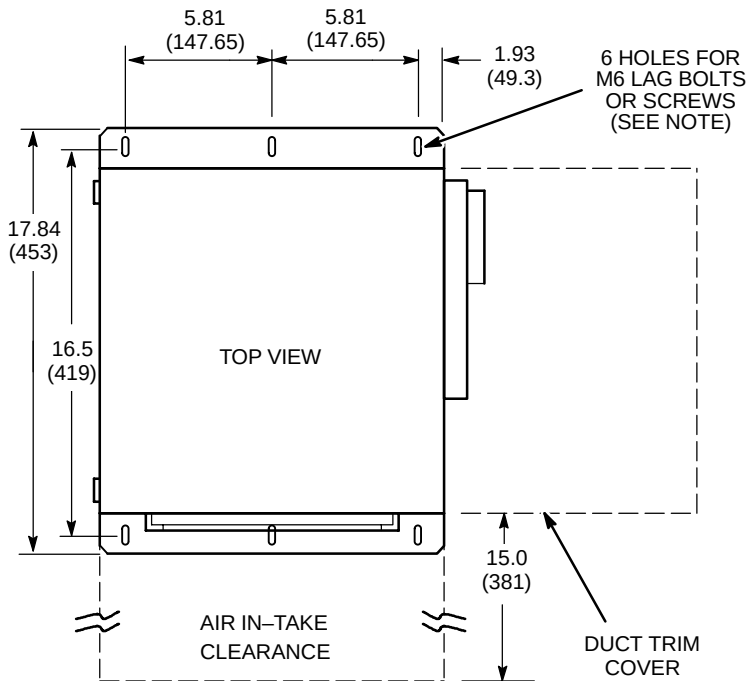
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 374 lbs (170 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INADEQUATE VENTILATION WILL CAUSE A REDUCTION IN COOLING CAPACITY AND, IN EXTREME CASES, COMPRESSOR FAILURE. SERVICING THIS UNIT MAY REQUIRE MOVEMENT OF CABINET WITHIN AIR FLOW CLEARANCE AREA.
- INDICATES CENTER OF GRAVITY 
- FOR 50 HZ UNIT ONLY: A SMALL TRANSFORMER IS LOCATED ON REAR OF THE CABINET (NOT SHOWN). ADDS 4.2 IN. (107 MM) TO DEPTH OF CABINET.

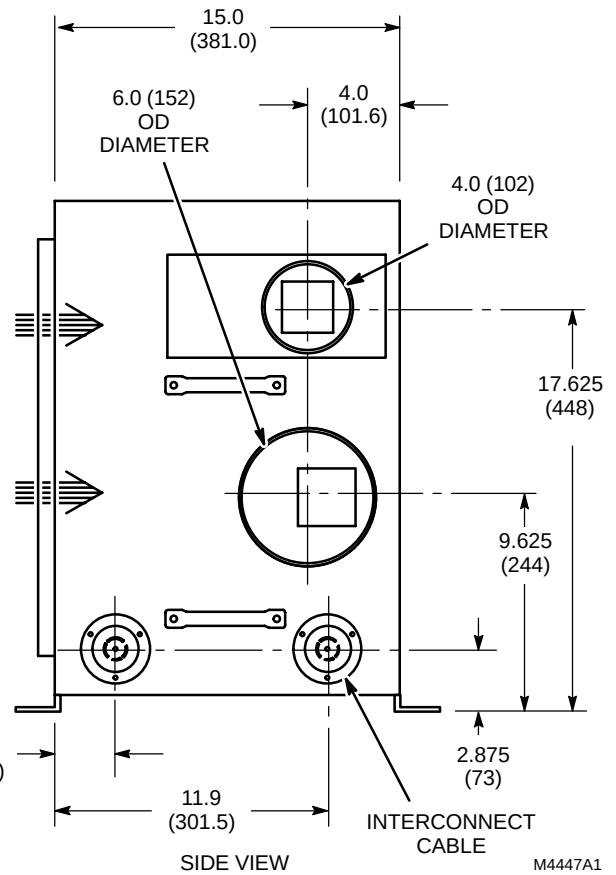


WATER CHILLER (WC1) FOR GRADIENT COIL COOLING WATER

ILLUSTRATION 2-25



FRONT VIEW



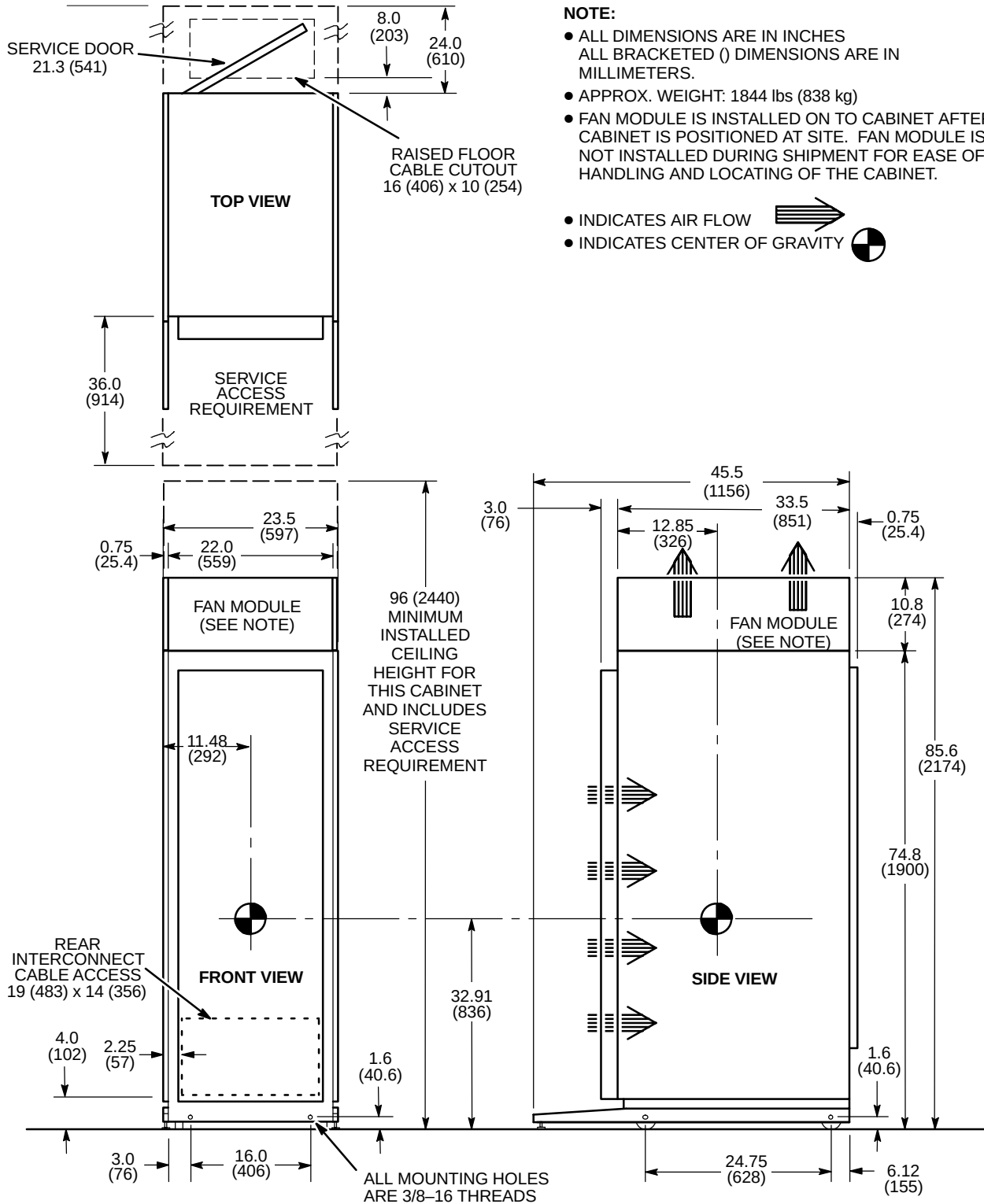
SIDE VIEW

BLOWER BOX (MG6)
ILLUSTRATION 2-26

NOTE:

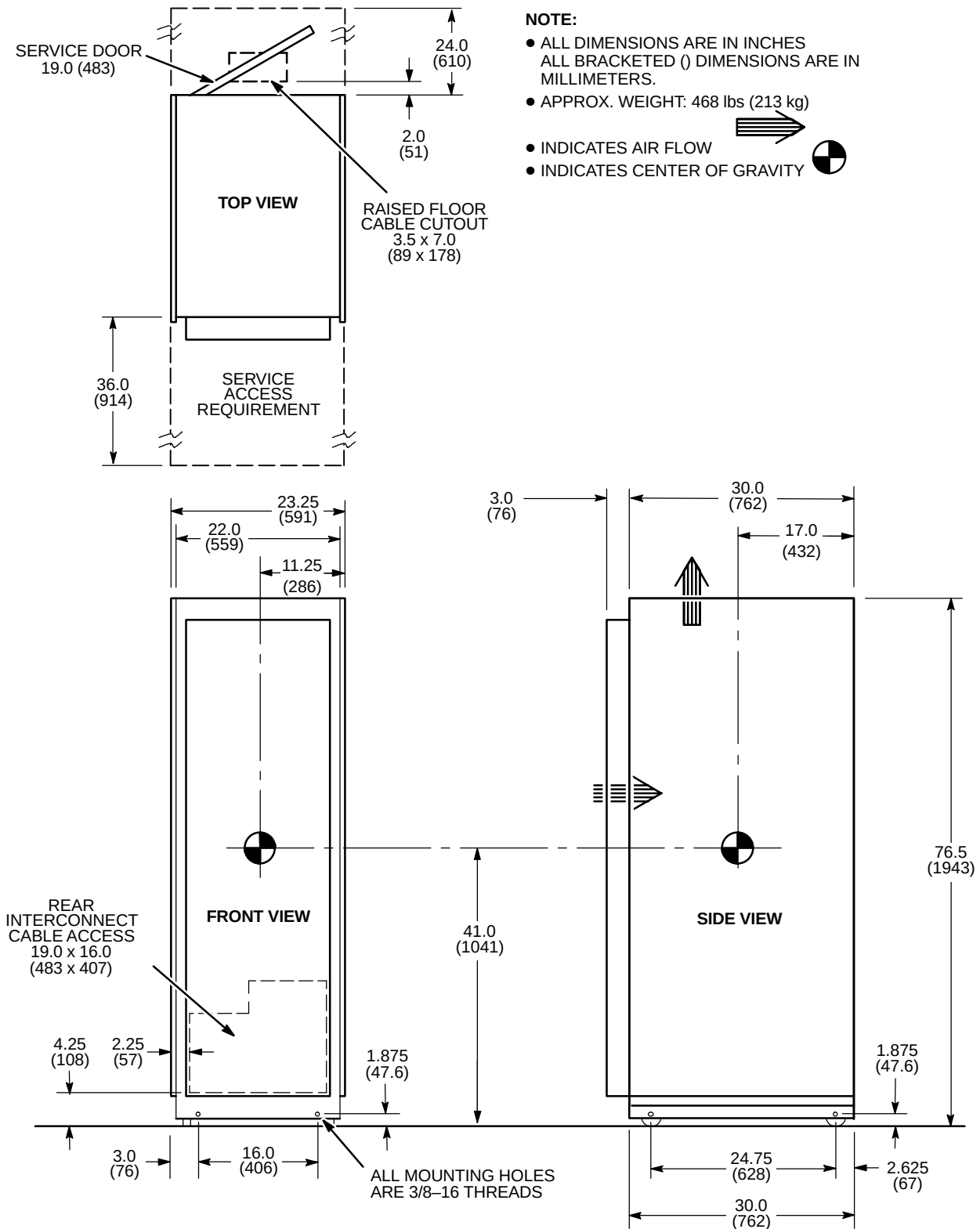
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 47 lbs (21 kg)
- BLOWER BOX MUST BE ANCHORED DUE FERROUS COMPONENTS.
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 

M4447A1



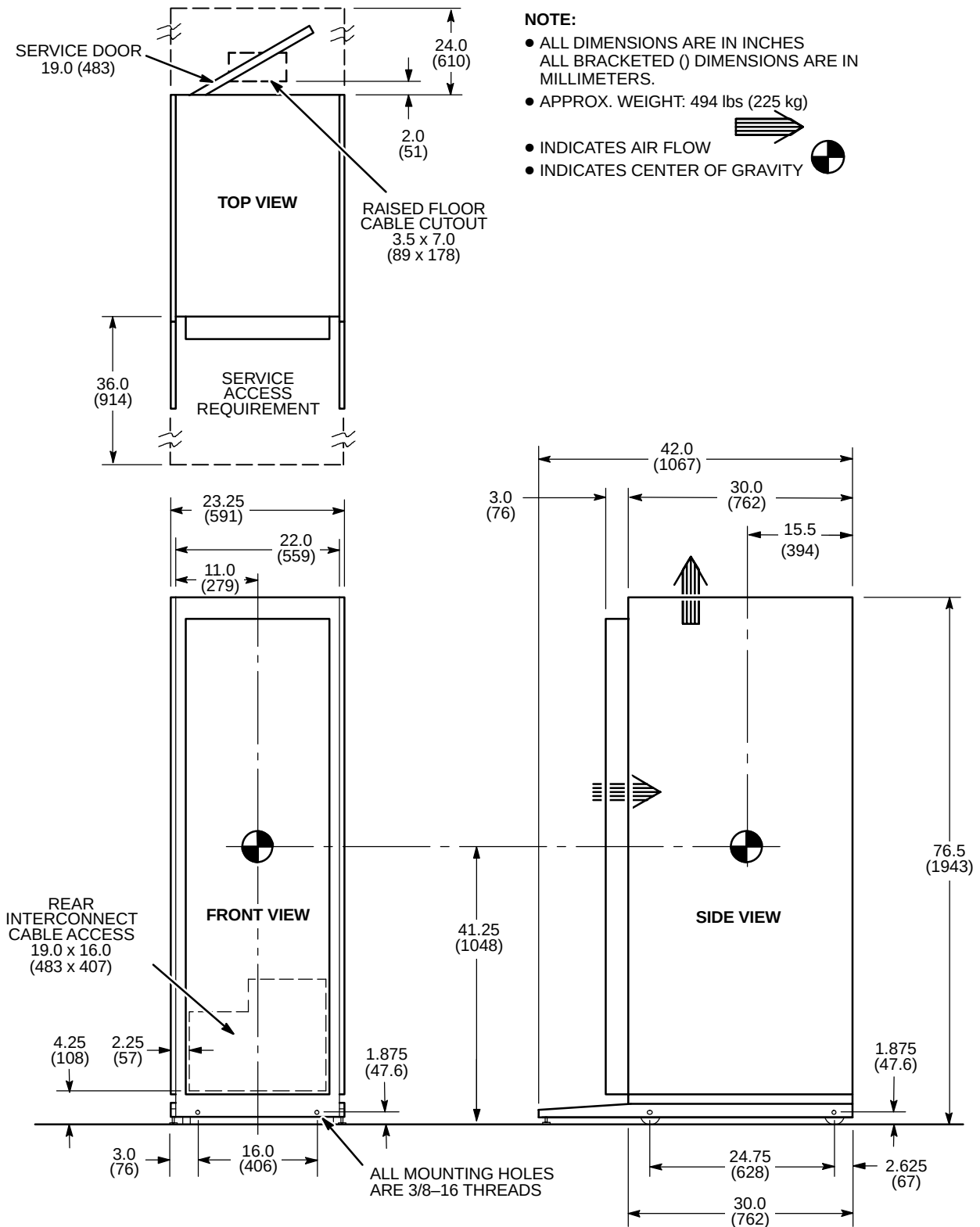
ACGD/PDU CABINET (MR3)

ILLUSTRATION 2-27



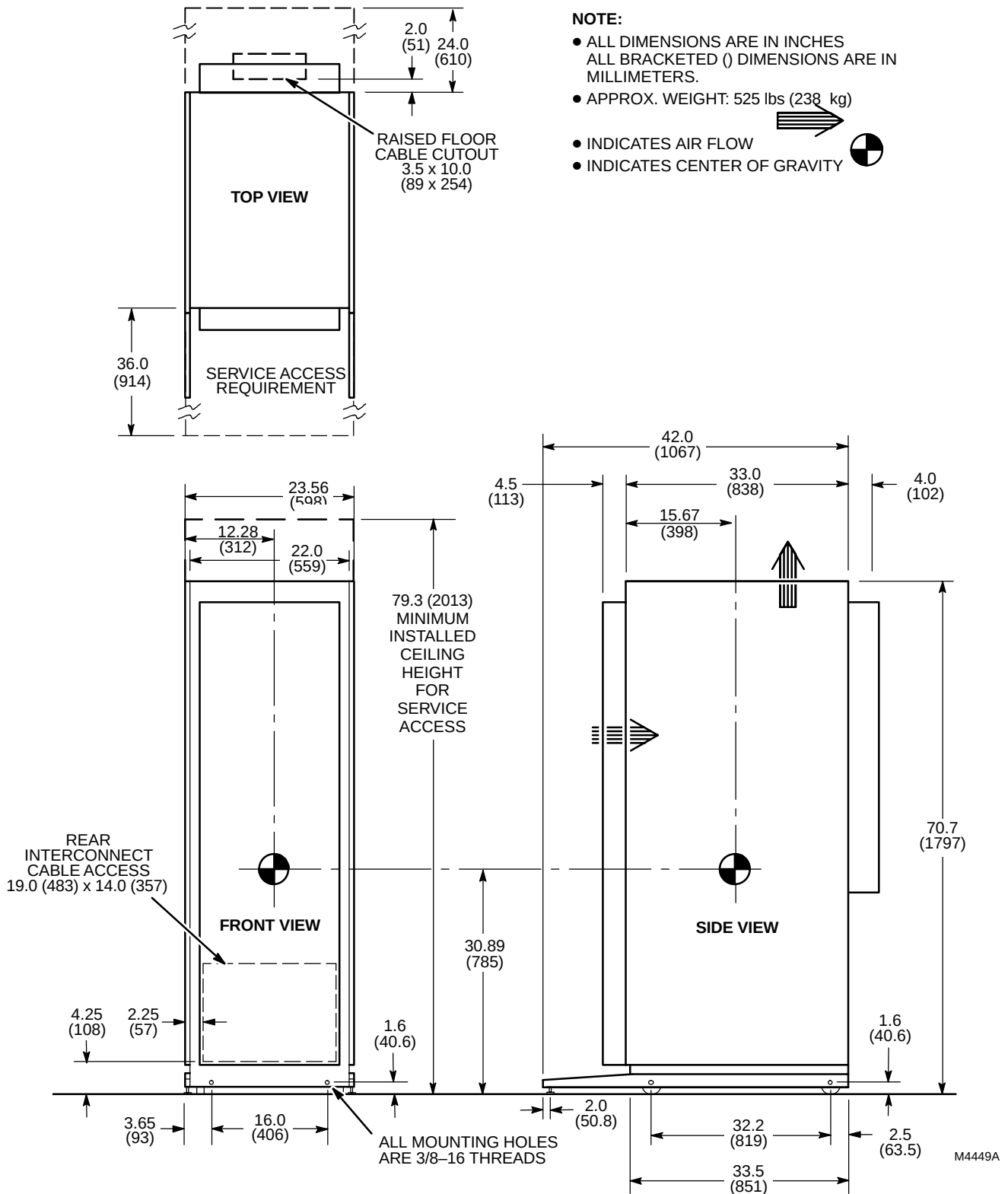
SYSTEM CABINET (MR2) FOR 1.5T SIGNA MR/i WITH EXCITE TECHNOLOGY

ILLUSTRATION 2-28



SYSTEM CABINET (MR2) FOR 1.5T SIGNA MR/i, 1.0T SIGNA MR/i, & 1.5T SIGNA CV/i

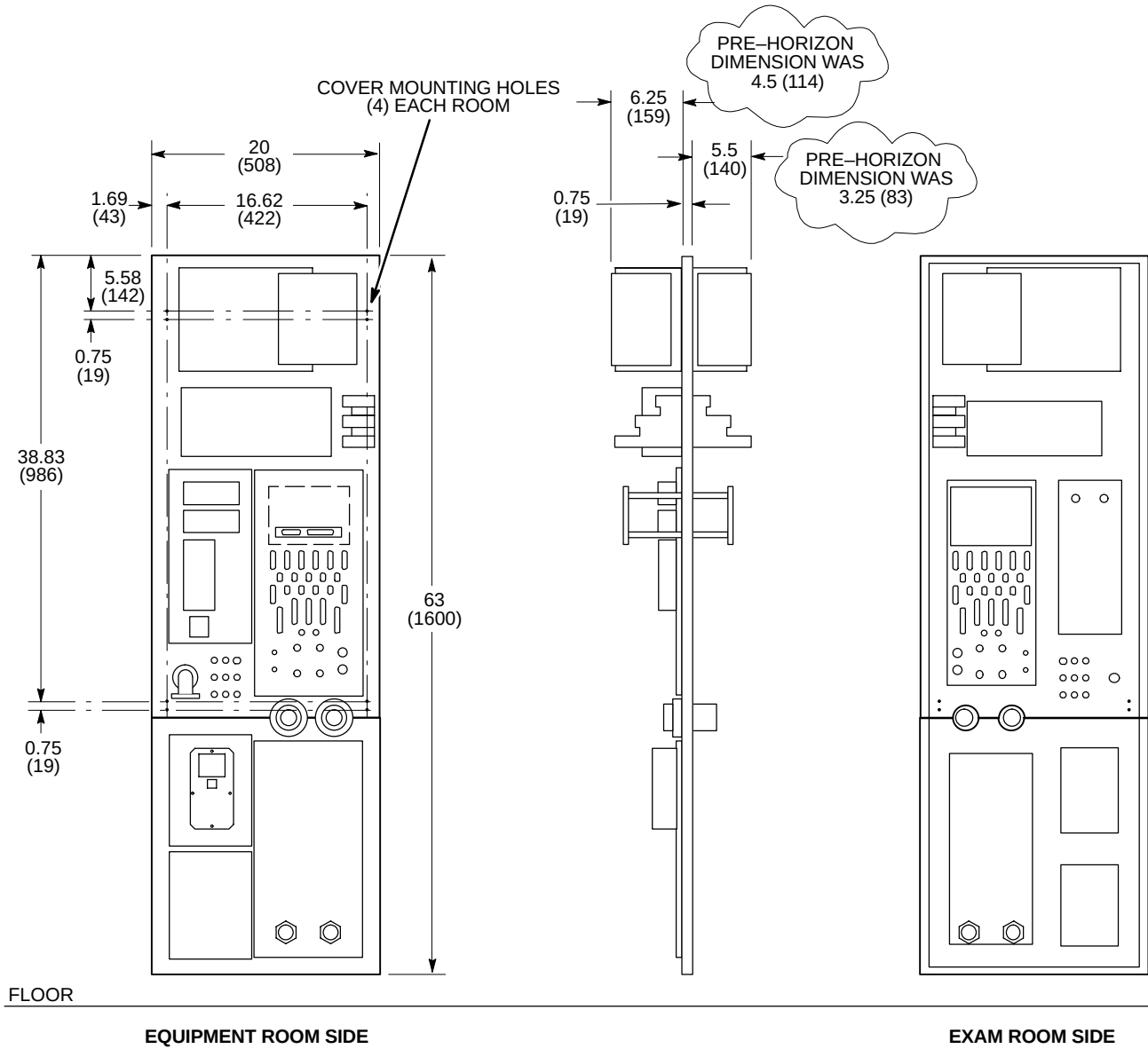
ILLUSTRATION 2-29



SRF CABINET (MR1)
ILLUSTRATION 2-30

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 52 lbs (23.6 kg)

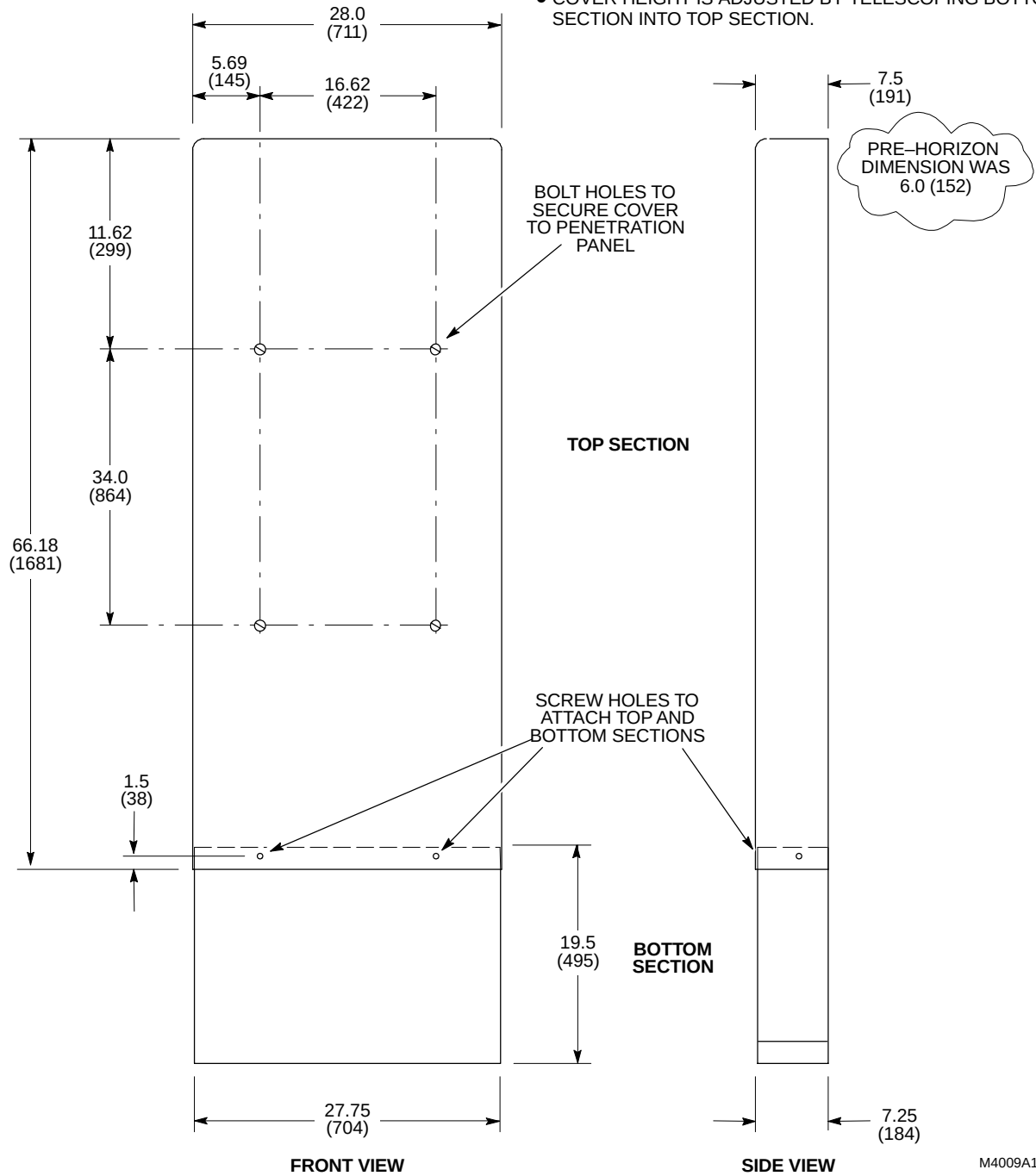


M4494A

PENETRATION PANEL (PP1)
ILLUSTRATION 2-31

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- COVER USED ON BOTH EXAM ROOM AND EQUIPMENT ROOM SIDES OF PENETRATION PANEL. TWO COVERS ARE PROVIDED.
- COVER HEIGHT IS ADJUSTED BY TELESOPING BOTTOM SECTION INTO TOP SECTION.

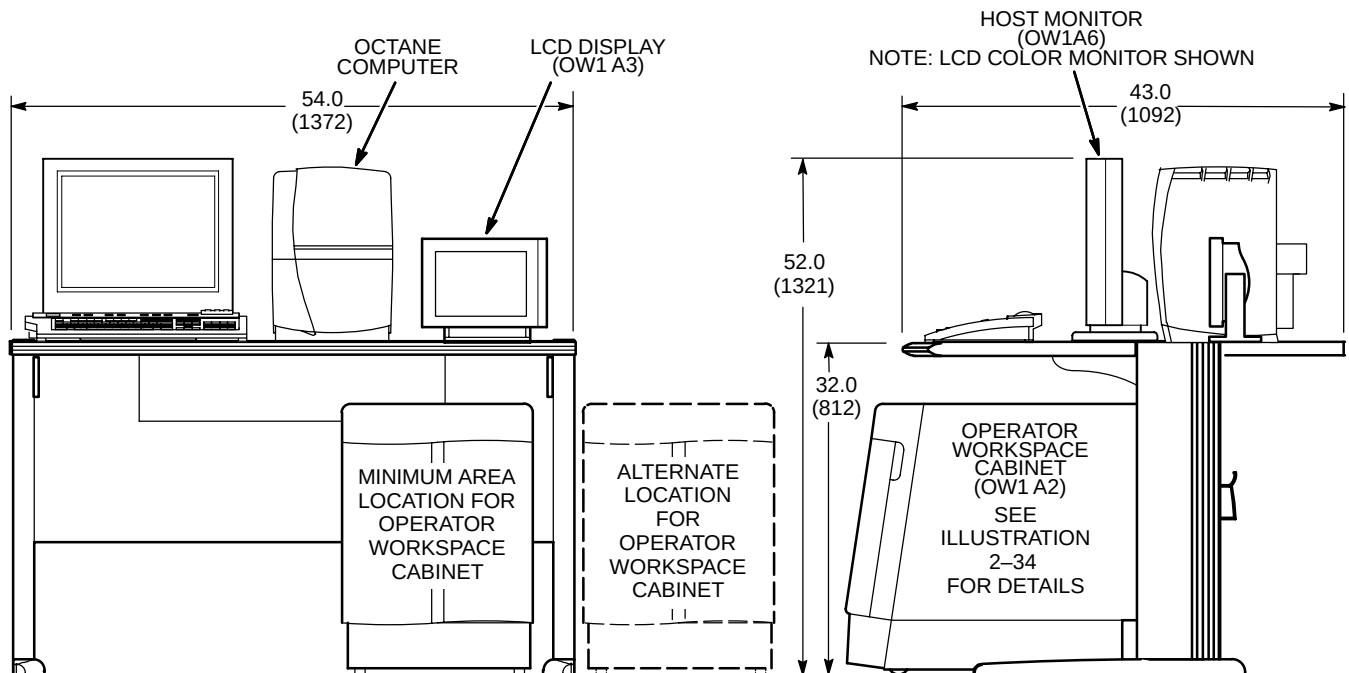
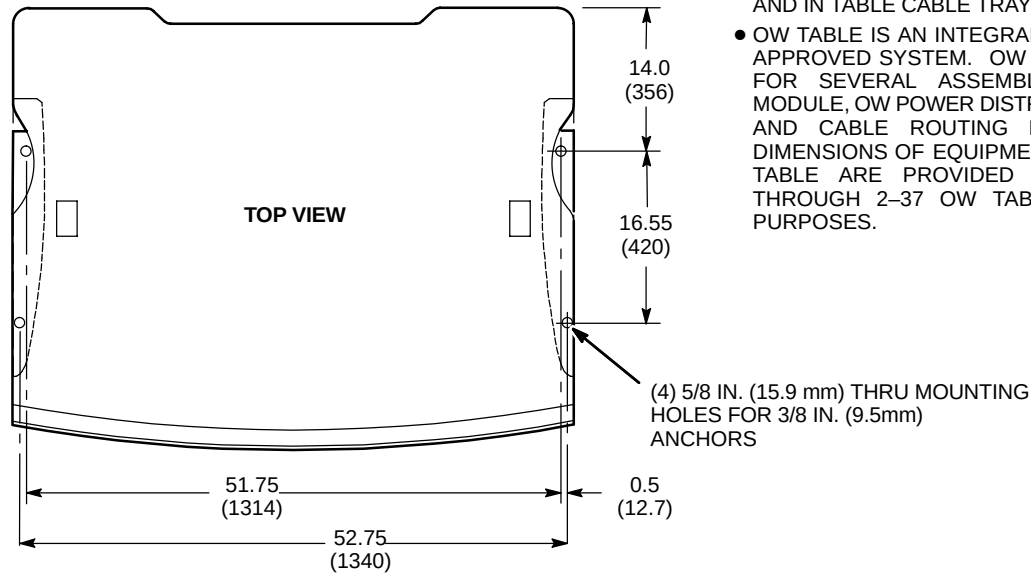


PENETRATION PANEL COVER
ILLUSTRATION 2-32

M4009A1M

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- ASSEMBLIES WHICH MOUNT TO UNDERSIDE OF TABLE AND IN TABLE CABLE TRAY ARE NOT SHOWN.
- OW TABLE IS AN INTEGRAL PART OF THE REGULATORY APPROVED SYSTEM. OW TABLE PROVIDES MOUNTING FOR SEVERAL ASSEMBLIES (E.G. OW INTERFACE MODULE, OW POWER DISTRIBUTION BOX, MODEM, DASM) AND CABLE ROUTING FOR OW INTERCONNECTS. DIMENSIONS OF EQUIPMENT LOCATED ON TOP OF OW TABLE ARE PROVIDED IN ILLUSTRATIONS NO TAG THROUGH 2-37 OW TABLE TOP LAYOUT PLANNING PURPOSES.



FRONT VIEW

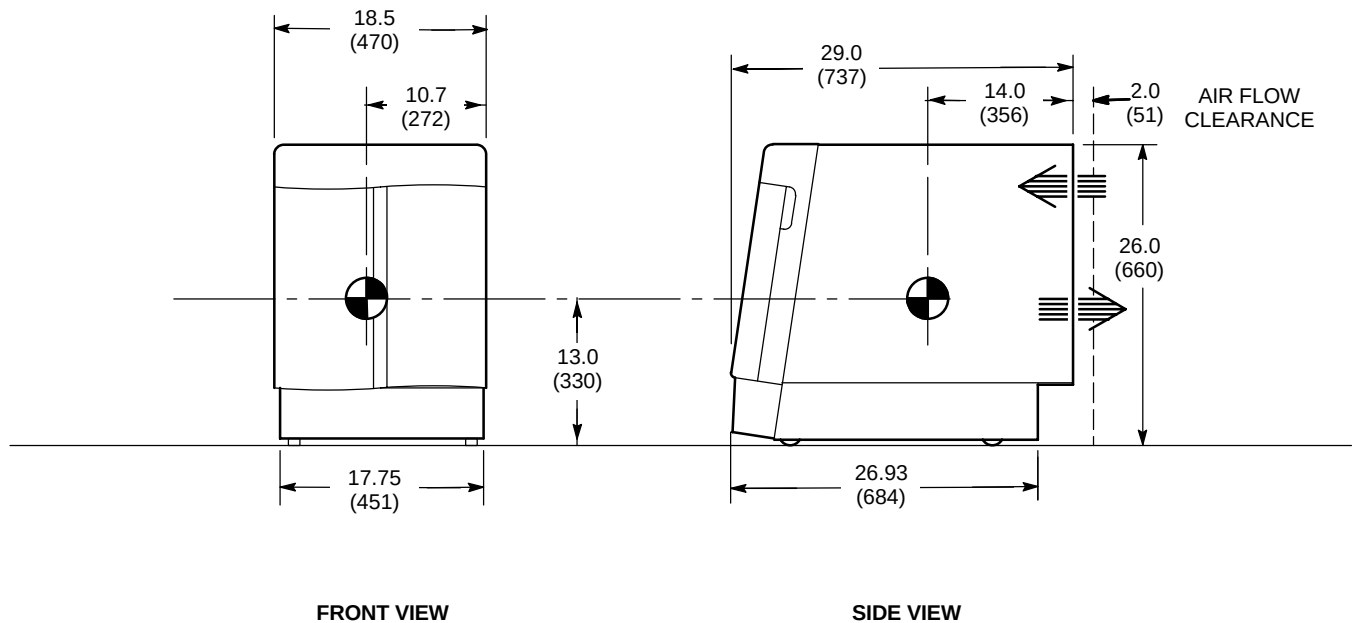
SIDE VIEW

OPERATOR WORKSPACE (OW1)

ILLUSTRATION 2-33

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 192 lbs (87 kg)
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 

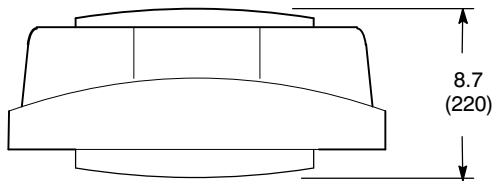


OPERATOR WORKSPACE CABINET (OW1 A2)

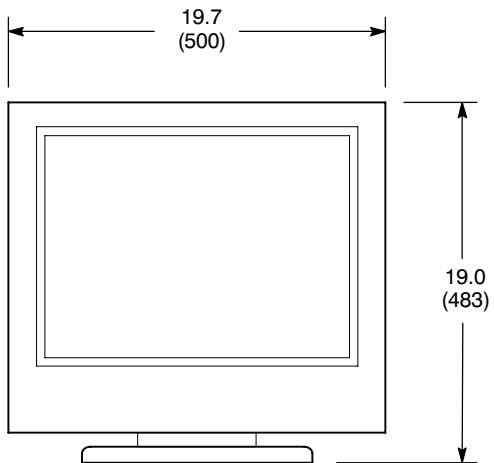
ILLUSTRATION 2-34

NOTE:

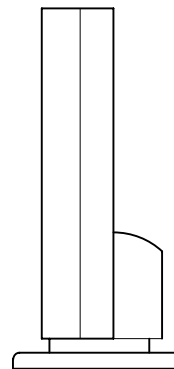
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



TOP VIEW



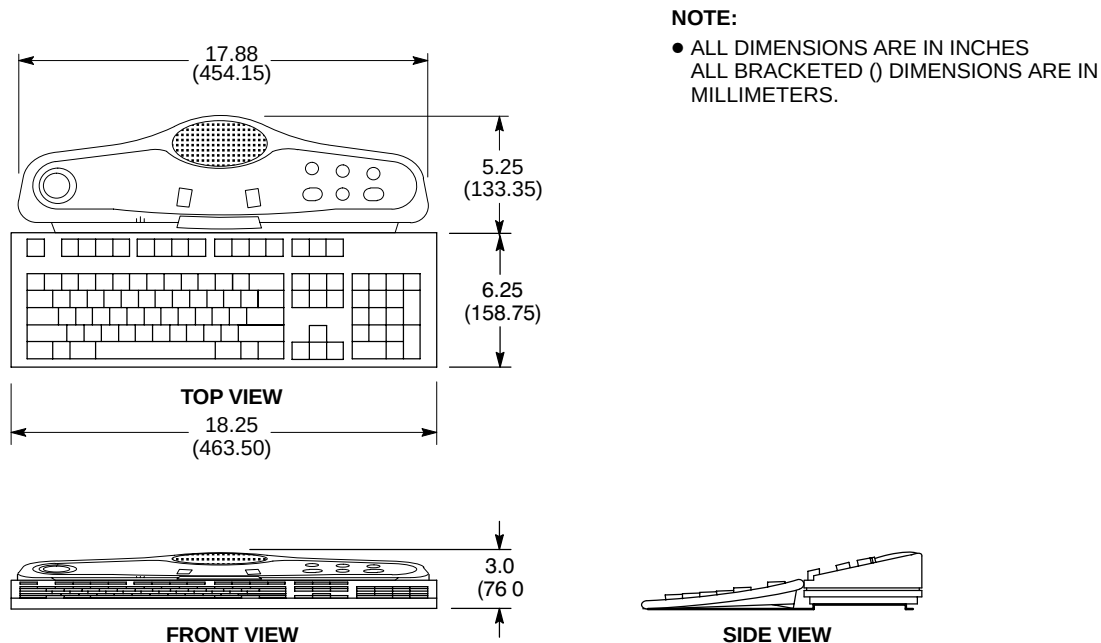
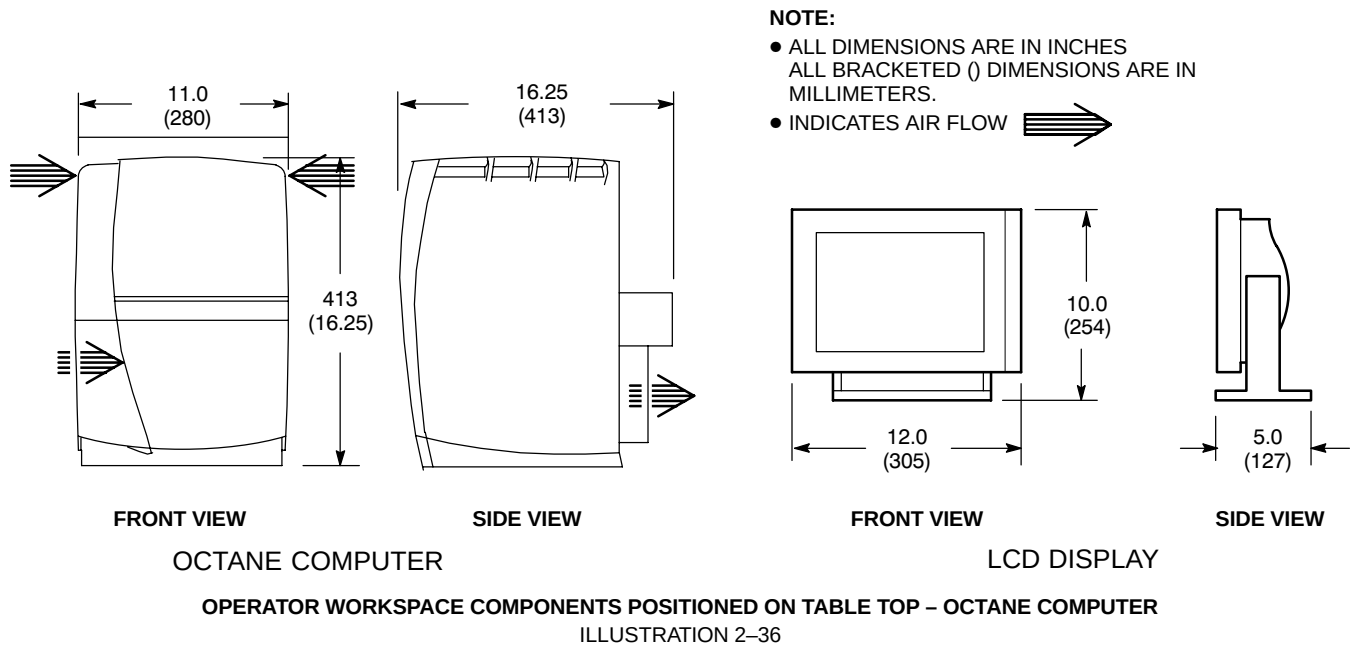
FRONT VIEW



SIDE VIEW

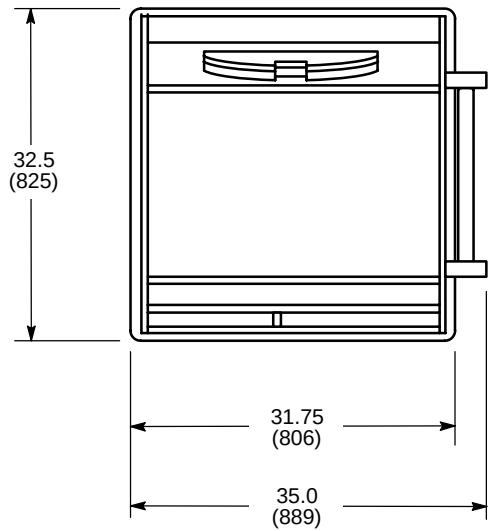
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR OPTION

ILLUSTRATION 2-35

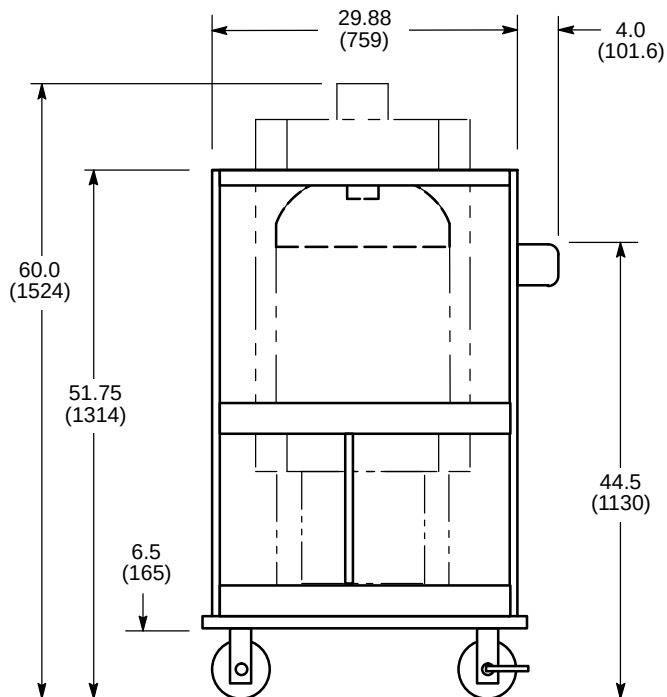


NOTE:

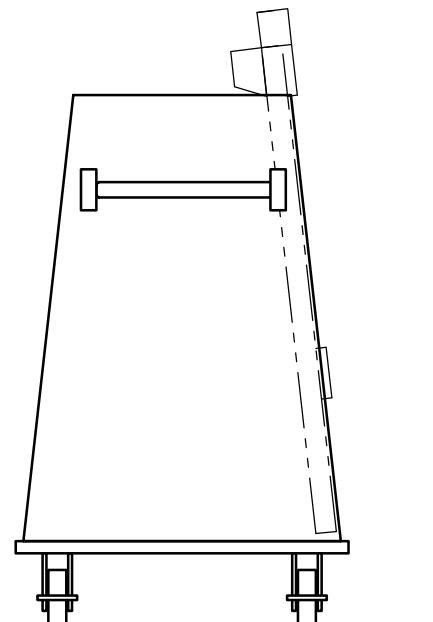
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT INCLUDING PHANTOMS: 350 lbs (159 kg)



TOP VIEW



FRONT VIEW

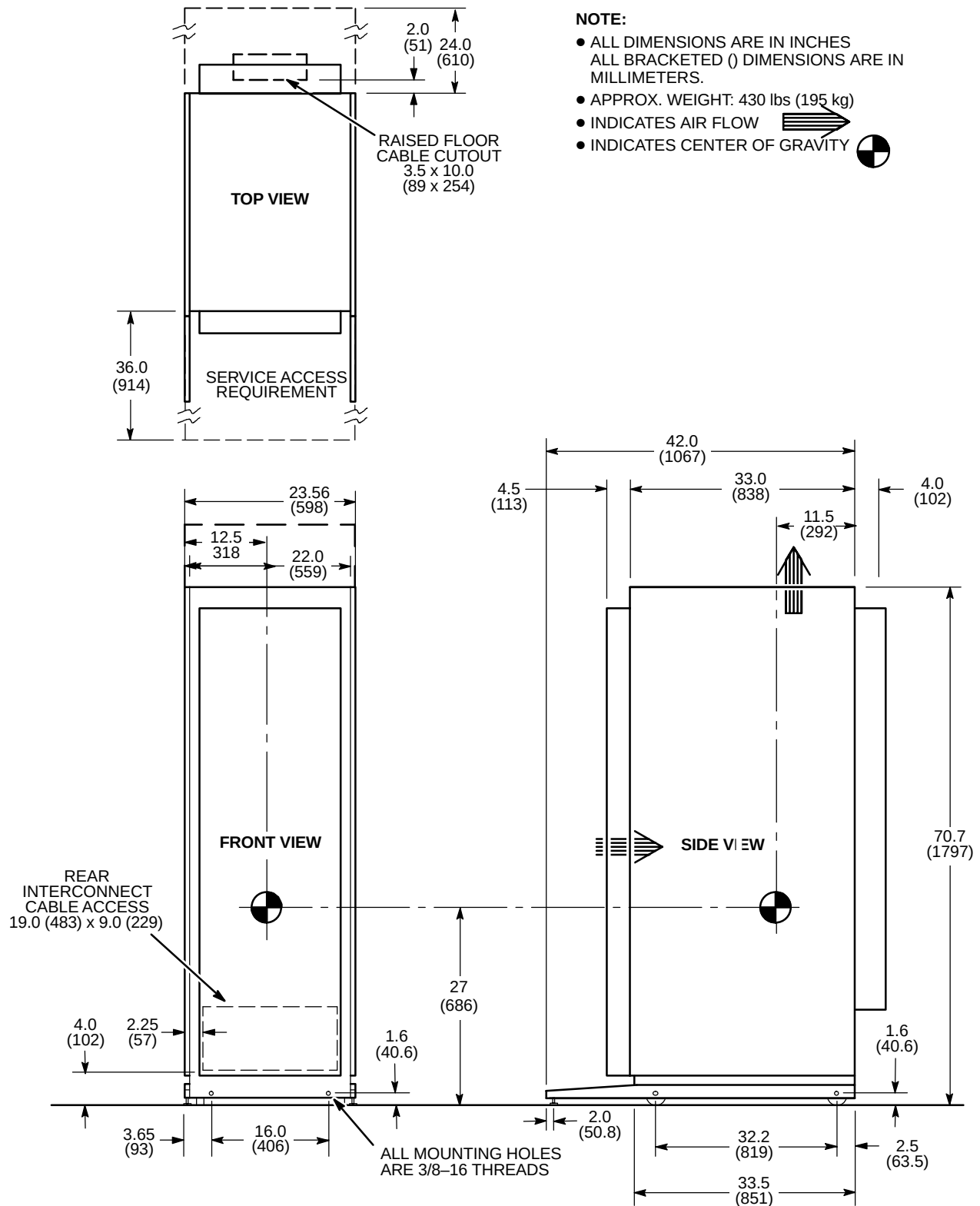


SIDE VIEW

M4454A1

SPT PHANTOM SET SHIPPING/STORAGE CABINET

ILLUSTRATION 2-38



MNS CABINET (MNS) - 1.5T OPTION

ILLUSTRATION 2-39